Chapter 11 - Cascading style sheets

With HTML tags, we can create a basic structure of a web page. Almost everything, from text to images can be created using these tags. But, these tags just provide a structure that is not appealing and attractive.

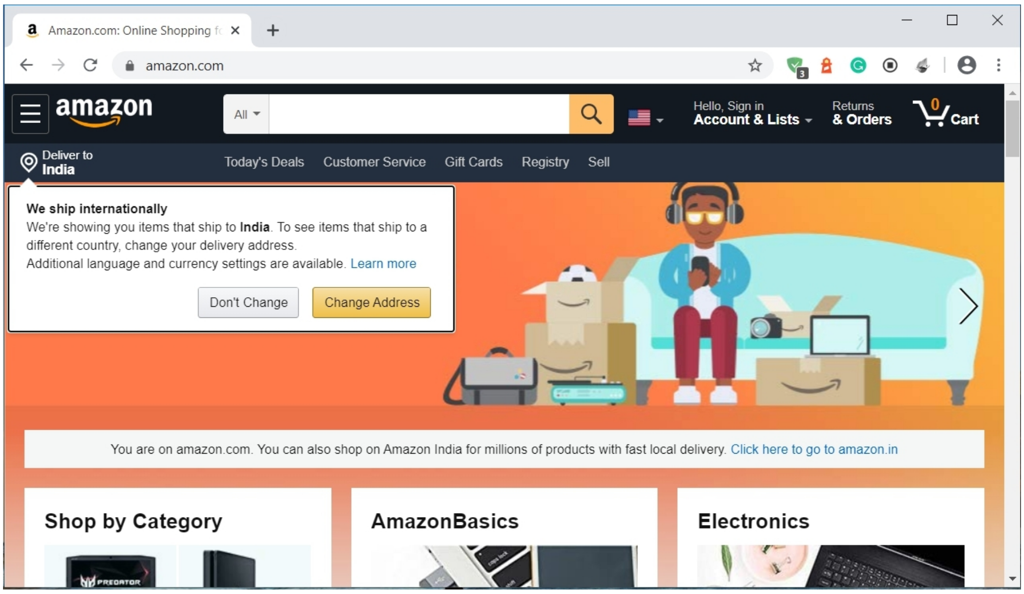
In the modern web, **websites are attractive and beautiful.** Everything is presented in a proper way with a lot of neatness and clarity. We can find text in different sizes, colors, or styles, and backgrounds with different colors and images, and many more. All of these **presentations are done using CSS.**

What is CSS and why do we use it?

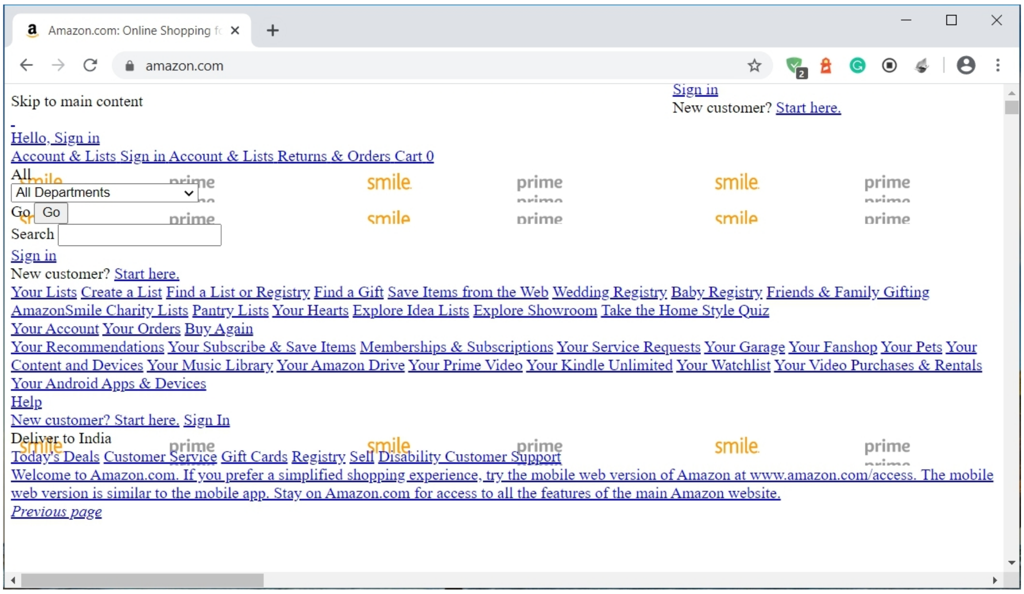
CSS stands for Cascading Style Sheets. It is another technology of the World Wide Web that is used to present an HTML document in a better way.

So basically, **CSS is a stylesheet language that is used with HTML for formatting.** It is another important part of web development because no website today is complete with CSS.

Have a look at amazon's homepage.



Now, observe it again, this time with no CSS.



It is understandable how important CSS is.

CSS not only turns a boring HTML page into an attractive one but also helps in placing the elements in proper positions.

Summary

- CSS is used to present an HTML document in a better and attractive way.
- CSS can be applied to any HTML element.

Chapter 12 - Syntax and ways of using CSS

Syntax of CSS

So let's start with the syntax. There are two parts in the CSS rule-set - **selector and declaration.**

CSS is applied to a particular HTML element or a group of HTML elements. But first, we need something to find these elements. **A selector in CSS is used to find the HTML elements.**

The declaration block has the CSS. **A single declaration has a key-value pair.** The key is the CSS property and it is separated from the value by a colon.

```
p {

    color : red;

}
```

Here, p is the selector that is selecting the <p> tags in the document. There are several types of CSS selectors and we will discuss all of them in a separate chapter.

The declaration block has one declaration in it, where "color" is the property and "red" is its value. We can also have multiple declarations and each declaration should be separated by a semicolon.

Ways of using CSS

There are three ways of using CSS - **External, Internal, and Inline.**

External CSS

When we have a lot of CSS and the CSS is common in multiple HTML documents, the external CSS is the right way. As the name suggests, the CSS is written in a **separate file with .css extension.**

```
p {

    color : red;

}

h1 {
    color : blue;
}
```

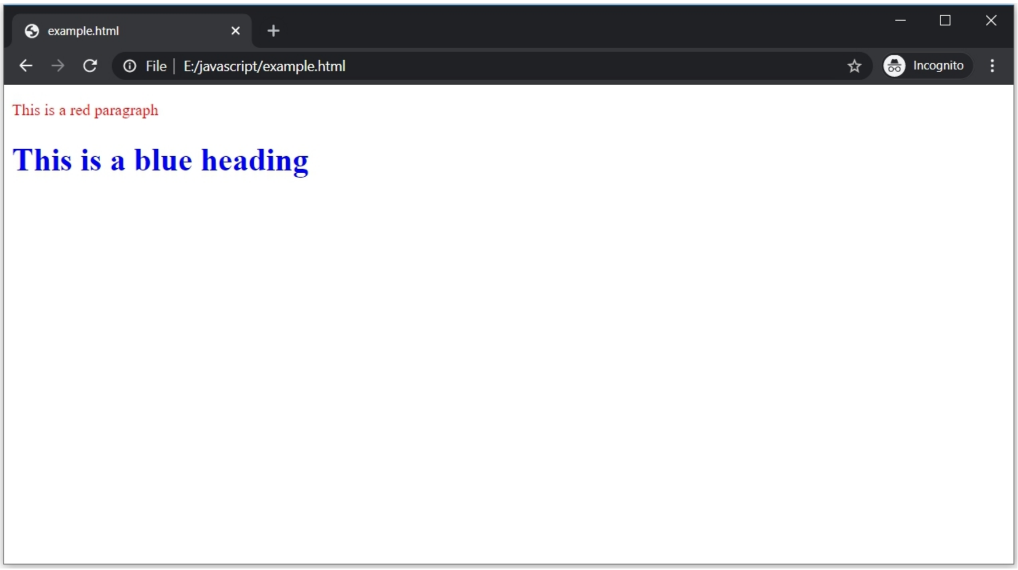
The name of this file is example.css. To use it in an HTML file, we need to include a **reference of it in the head section of the HTML using the <link> tag.**

```
<!DOCTYPE html>
<html>
    <head>
        <link rel="stylesheet" type="text/css" href="example.css">
    </head>
<body>

    <p>This is a red paragraph</p>

    <h1>This is a blue heading</h1>

</body>

</html>
```

Observe the <link> tag. It has three mandatory attributes:

- rel: Specifies the relationship between the linked file and the current file. As the linked file is a CSS file, we have to specify "stylesheet" as its value.
- type: Specifies the media type of the file that is linked with HTML file. The type of a CSS file is "type/CSS".
- href: Specifies the location of the linked file.

This is how the <link> tag is used to link the external CSS file with an HTML file.



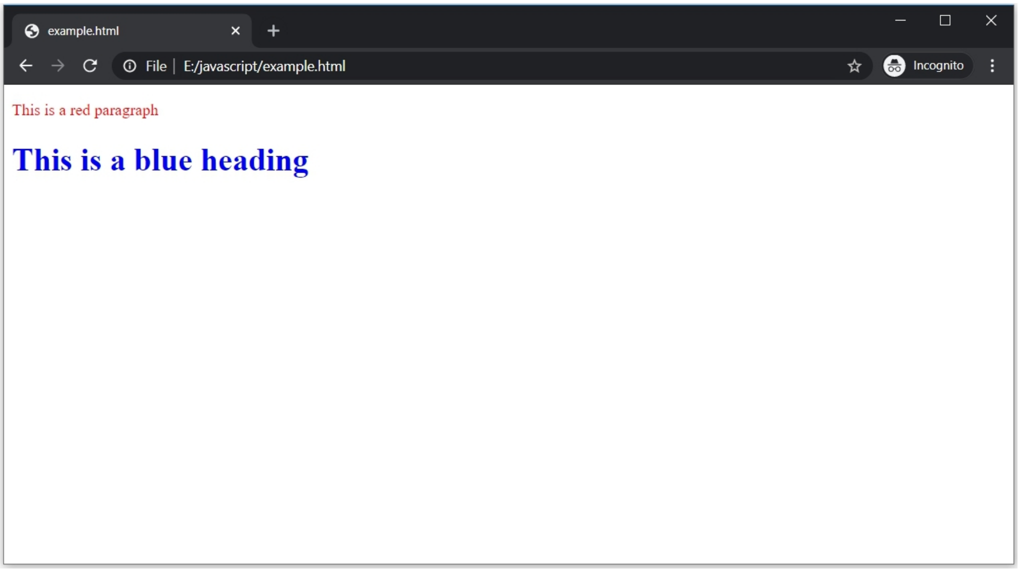
The color of the paragraph and heading is changed to red and blue, respectively.

Internal CSS

So the external CSS should be preferred if CSS is common for multiple pages. But if there is a unique CSS for a single page, use the internal CSS.

In this way, **the CSS is placed inside the <style> tag** which, in turn, is placed inside the head section.

```html
<!DOCTYPE html>
<html>
    <head>
        <style>
            p {
                color : red;
            }

            h1 {
                color : blue;
            }
        </style>
    </head>
<body>

    <p>This is a red paragraph</p>

    <h1>This is a blue heading</h1>

</body>

</html>
```

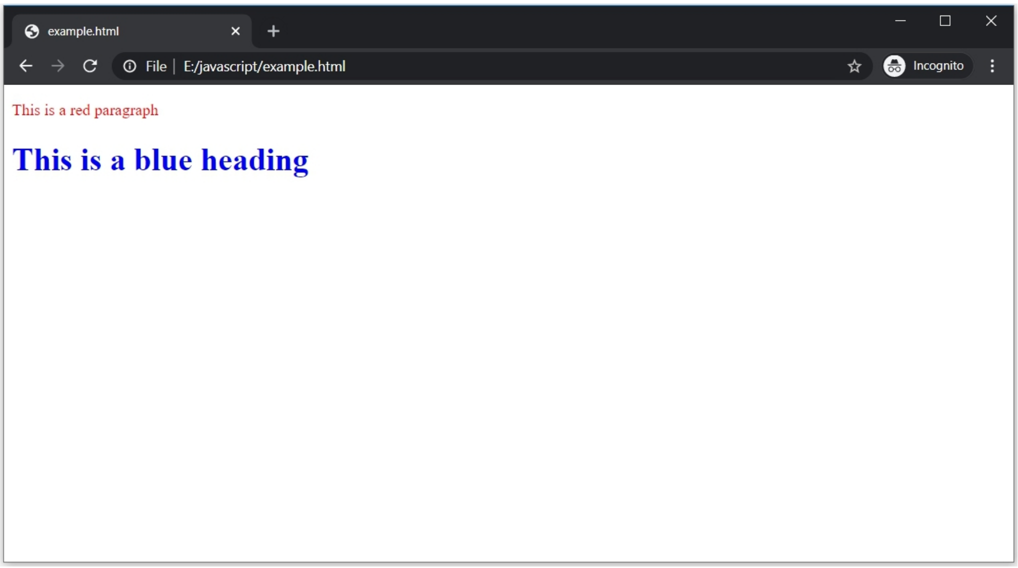


Inline CSS

The third way is a bit different. **Every HTML tag has an attribute known as the style attribute.** This attribute is used to specify CSS for a single tag.

```
<!DOCTYPE html>
<html>
<body>

    <p style="color : red;">This is a red paragraph</p>

    <h1 style="color : blue;">This is a blue heading</h1>

</body>

</html>
```

There are no selectors in inline style.



Priority order

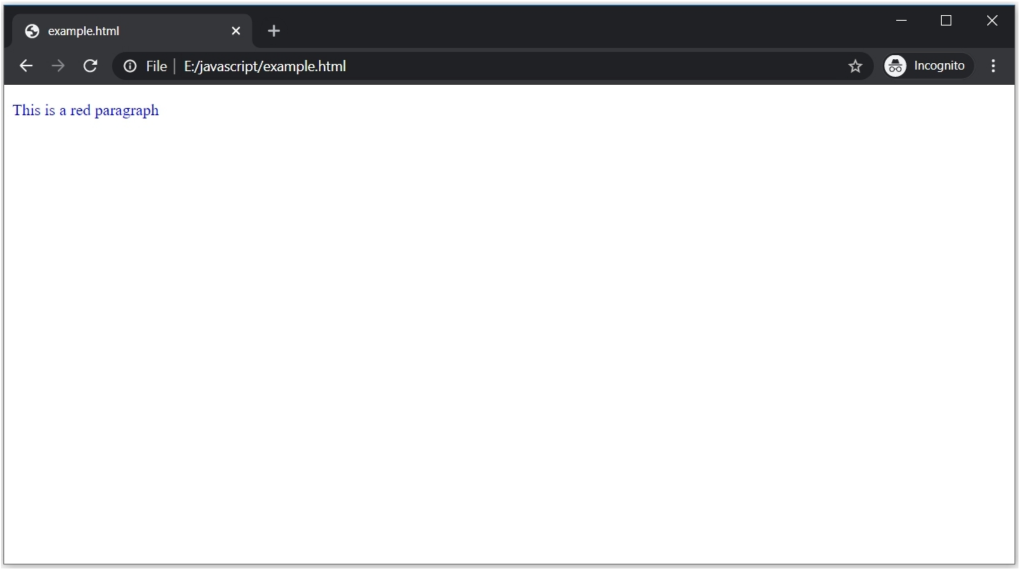
What if the same CSS is used for the same HTML element in a single HTML file with all the three ways? For example, observe the following HTML file.

```html
<!DOCTYPE html>
<html>
    <head>
        <link rel="stylesheet" type="text/css" href="example.css">

        <style>
            p {
                color : green;
            }
        </style>


    </head>
<body>

    <p style="color : blue;">This is a red paragraph</p>

</body>

</html>
```

There is a single <p> tag and internal as well as inline CSS is applied on it. Also, example.css is linked with it.

```css
p {

    color : red;

}
```

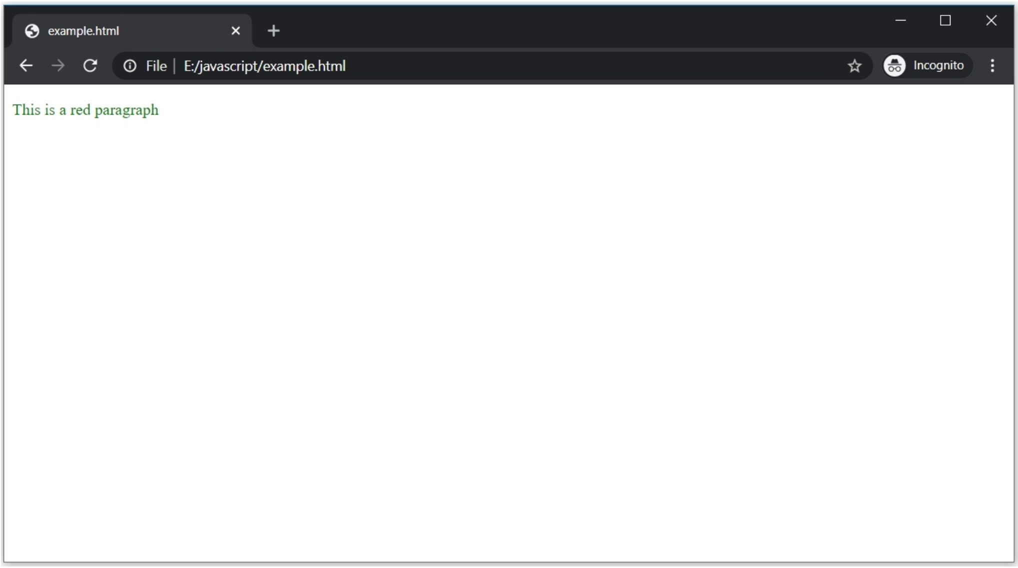
The external CSS is also targeting the same property of the <p> tag. But, the value for the color is different in all the cases; "blue" in inline, "green" in internal, and "red" in external. What do you think will happen? What will be the color of the <p> tag? Let's see.



The color is blue, this means **inline CSS has more priority over the other two.** Now, let's remove the inline CSS and see, who has more priority among the internal and external CSS.

```
<!DOCTYPE html>
<html>
    <head>
        <link rel="stylesheet" type="text/css" href="example.css">

        <style>
            p {
                color : green;
            }
        </style>


    </head>
<body>

    <p>This is a red paragraph</p>

</body>

</html>
```

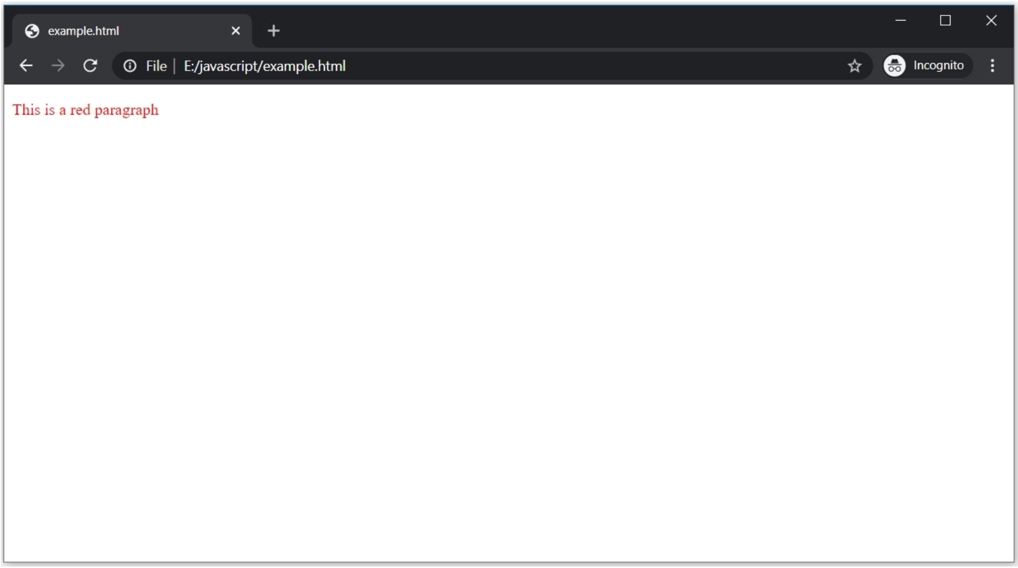
Let's see what is the color of the <p> tag now.



The color is green, so it means the internal CSS has priority over the external CSS, right? Let's interchange the order their placement inside the <head> tag.

```html
<!DOCTYPE html>
<html>
    <head>
        <style>
            p {
                color : green;
            }
        </style>


        <link rel="stylesheet" type="text/css" href="example.css">


    </head>
<body>

    <p>This is a red paragraph</p>

</body>

</html>
```

This time, internal CSS is placed before the external CSS.



The color of the <p> tag is red. So, **the priority among the internal and external CSS depends upon the order in which they are placed inside the head section.** The CSS which is placed later in the head section has more priority.

Summary

- There are two parts in CSS rule-set - selector and declaration.
- The selector is used to select an HTML element(s) while the declaration has the CSS properties and their corresponding values.
- A single declaration has a property and its value, separated by a colon.
- There can be multiple declarations in a block, each separated by a semicolon.
- Three ways of using CSS are external, internal, and inline.
- Externally, CSS is written in a separate file with .css extension. The reference to the CSS file is written using the <link> tag that is placed inside the head section.
- The internal CSS is written using the <style> tag inside the head section.
- Every HTML tag has a style attribute and this attribute is used for the inline CSS.
- The inline CSS has the priority over other two.
- The priority among the external and internal CSS depends on the order of their placement inside the head section. The latter placed will be given higher priority.

Chapter 13: CSS selectors

In the last chapter, we discussed the syntax of CSS. **Selectors are one of two parts of the CSS rule-set.** We discussed how to select an HTML element using their tag name. For example, the following CSS example selects all the <p> tag in the HTML file.

```
p {

    color : red;

}
```

Now, it does not matter, if there is only a single <p> tag in the HTML file or there hundreds, this CSS will be applied to all of them. But we do not want to apply the same CSS to all the paragraphs, right? Maybe, we want red color for one paragraph and blue for another. Similarly, there can be various other scenarios where CSS is needed only on some elements. There are different selectors for such scenarios.

Id selector

Observe the following HTML code.

```
<!DOCTYPE html>
<html>
    <head>


    </head>
    <body>

        <p>This is a red paragraph</p>
        <p>This is a blue paragraph</p>
        <p>This is a green paragraph</p>

    </body>

</html>
```

There are three paragraphs but no CSS is applied to them. We have to give different colors to each paragraph. How can we do it? We need to select a particular paragraph and apply a particular color to it. To do this, **the id selector is used.**

Every element has an "id" attribute. Id given to an element should be unique.

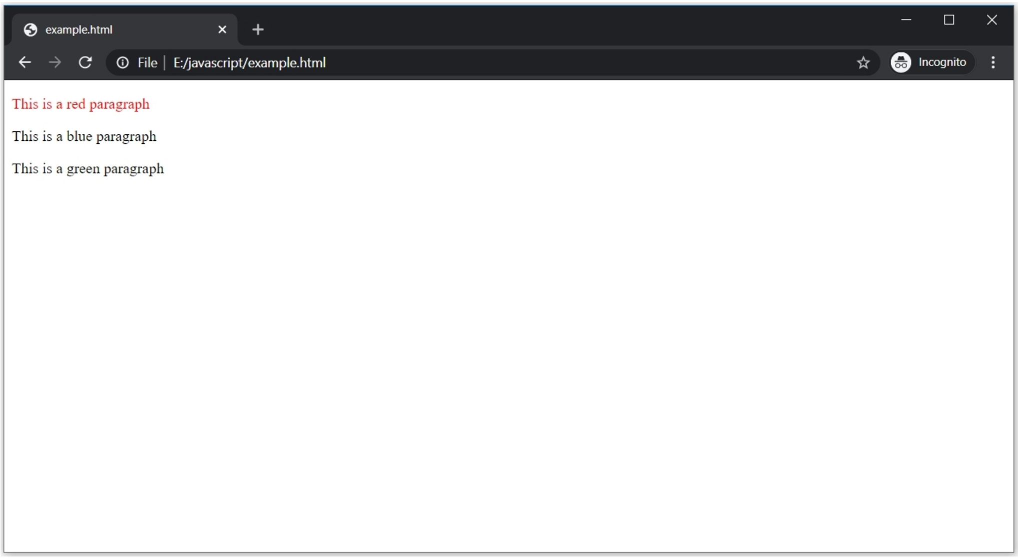
```
<!DOCTYPE html>
<html>
    <head>


    </head>
    <body>

        <p id="redp">This is a red paragraph</p>
        <p id="bluep">This is a blue paragraph</p>
        <p id="greenp">This is a green paragraph</p>

    </body>

</html>
```

Each paragraph has a unique id and unique CSS can be applied to each of them using this id.

To select an element using the id, **use the hash (#) sign followed by the id name.**

```
<!DOCTYPE html>
<html>
    <head>
        <style>
            #redp {
                color : red;
            }
        </style>
    </head>
    <body>

        <p id="redp">This is a red paragraph</p>
        <p id="bluep">This is a blue paragraph</p>
        <p id="greenp">This is a green paragraph</p>

    </body>

</html>
```

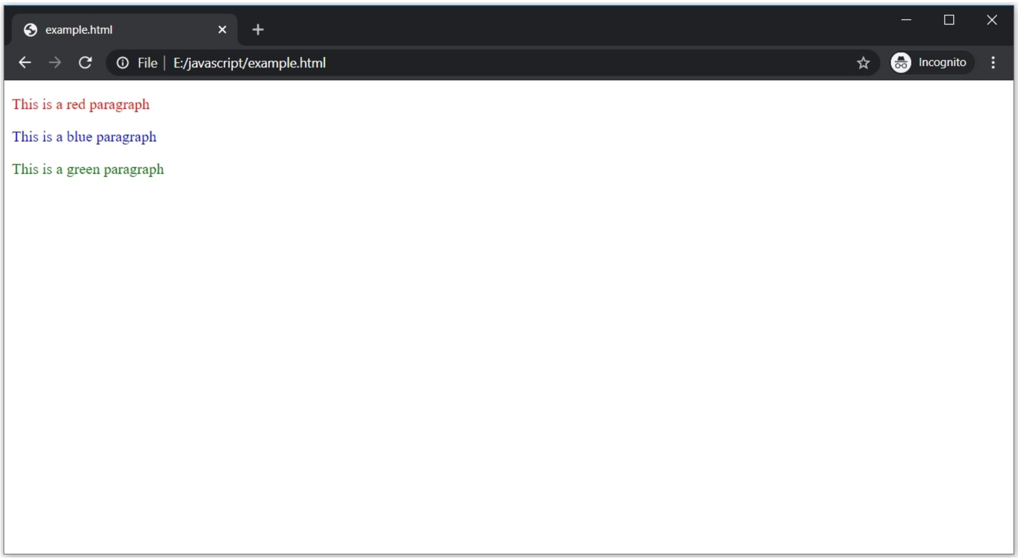
The CSS is applied to the paragraph whose id is "redp".



The color of the paragraph with id "redp" is changed to red. Other are not effects.

This is how the id selector is used to apply unique CSS to a particular element. Let's give their respective colors to other paragraphs.

```
<!DOCTYPE html>
<html>
    <head>
        <style>
            #redp {
                color : red;
            }

            #bluep {
                color : blue;
            }

            #greenp {
                color : green;
            }
        </style>
    </head>
    <body>

        <p id="redp">This is a red paragraph</p>
        <p id="bluep">This is a blue paragraph</p>
        <p id="greenp">This is a green paragraph</p>

    </body>

</html>
```



Remember, **an id can never start with a number.**

class selector

The id is usually unique for an element. But what if we need the same CSS for a group of elements? Not all, but some of them. Observe the following HTML file.

```
<!DOCTYPE html>
<html>
    <head>


    </head>
    <body>

        <p>This is a red paragraph</p>
        <p>This is a red paragraph</p>
        <p>This is a red paragraph</p>

        <p>This is a paragraph</p>
        <p>This is a paragraph</p>
        <p>This is a paragraph</p>


    </body>

</html>
```

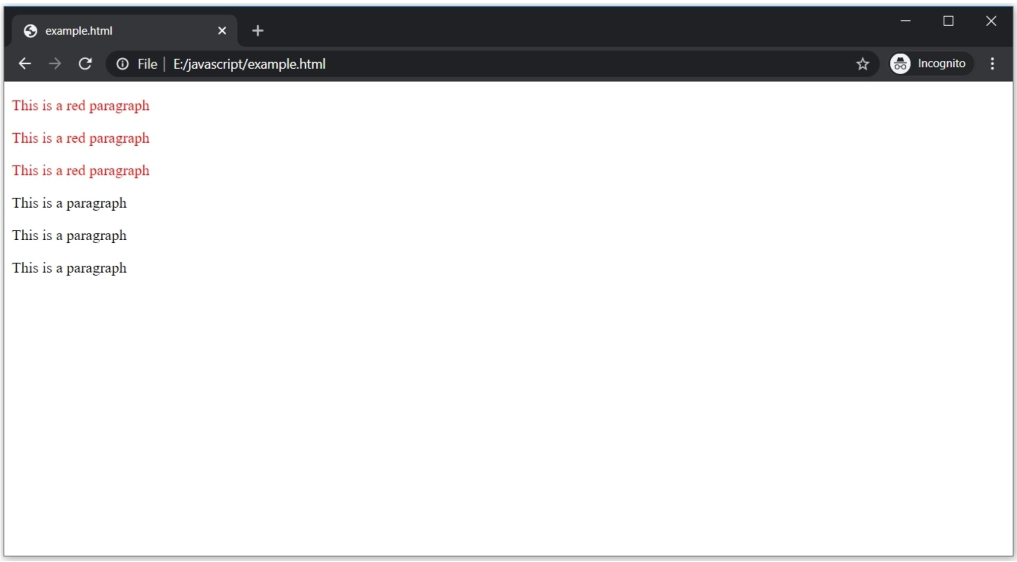
Out of all the six paragraphs, first, three should be red in color. We can give the same ids to them and it will work. But it is not recommended at all. For such cases, we have another attribute, which is known as the "class" attribute.

Grouping selector

The class selector is used to give the same CSS to multiple elements. To use this type of selector, use a dot (.) followed by the class name.

The same class is assigned for the first three <p> tags.

```
<!DOCTYPE html>
<html>
    <head>
        <style>
            .redp {
                color : red;
            }
        </style>


    </head>
    <body>

        <p class="redp">This is a red paragraph</p>
        <p class="redp">This is a red paragraph</p>
        <p class="redp">This is a red paragraph</p>

        <p>This is a paragraph</p>
        <p>This is a paragraph</p>
        <p>This is a paragraph</p>


    </body>

</html>
```

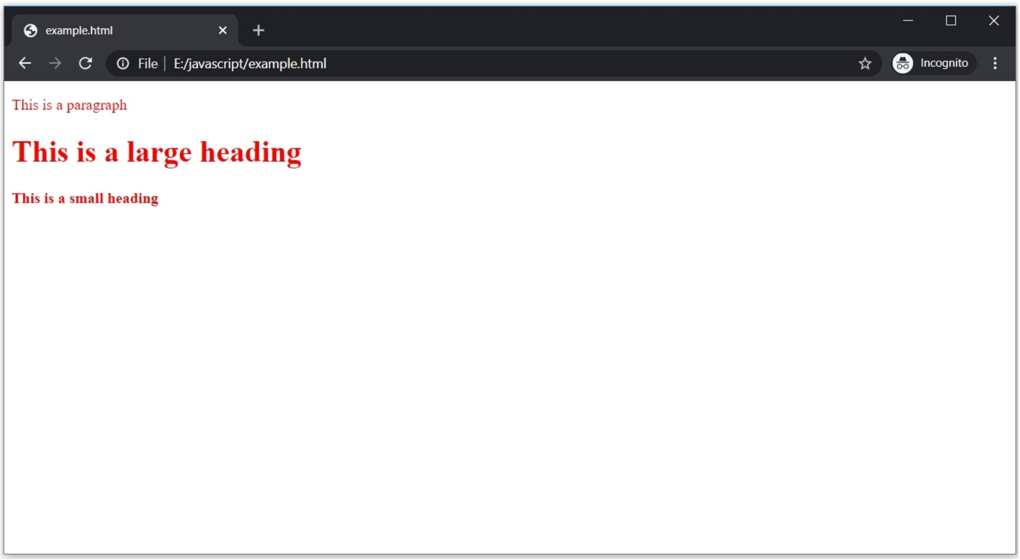


The CSS is applied to the first three paragraphs, while others are unaffected.

Like id, **a class name can also not start with a number.**

Now, suppose there are few elements with have the same definition and we want the same CSS for them.

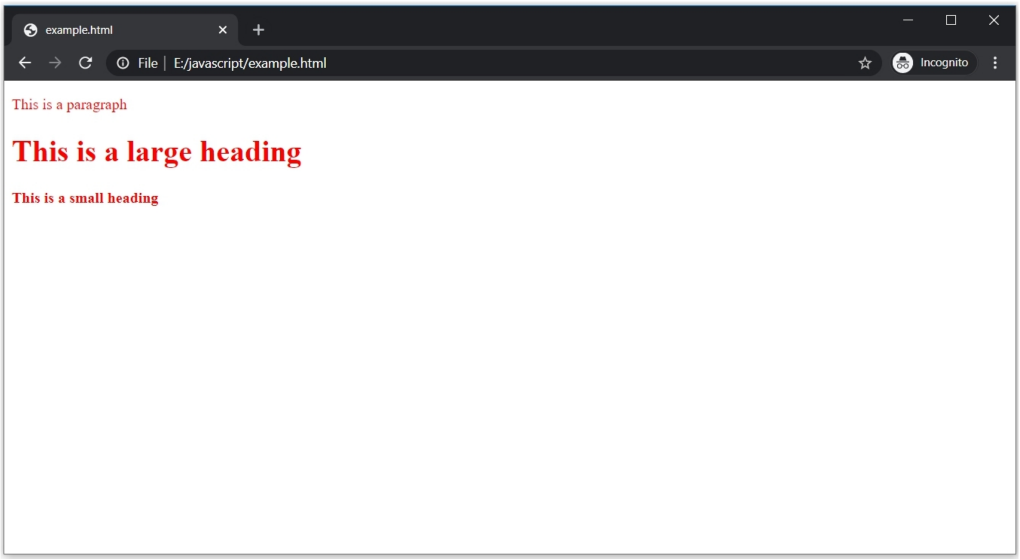
```
<!DOCTYPE html>
<html>
    <head>
        <style>
            p {
                color : red;
            }

            h1 {
                color : red;
            }

            h4 {
                color : red;
            }
        </style>


    </head>
    <body>

        <p> This is a paragraph</p>

        <h1> This is a large heading</h1>

        <h4> This is a small heading</h4>


    </body>

</html>
```



The <p>, <h1>, <h4> tag have same definitions. Observe the code above. Each of them has the same CSS defined separately. But why write separate code for each of them when the CSS is the same? We can use the class selector, but the class selector should be used for the same type of HTML element. In such a case, use the CSS grouping selector.

The CSS grouping selector is used to group different elements so that the same CSS can be applied to them. Just **use the element names and separate them with comma.**

```
<!DOCTYPE html>
<html>
    <head>
        <style>
            p, h1, h4 {
                color : red;
            }
        </style>


    </head>
    <body>

        <p> This is a paragraph</p>

        <h1> This is a large heading</h1>

        <h4> This is a small heading</h4>


    </body>

</html>
```



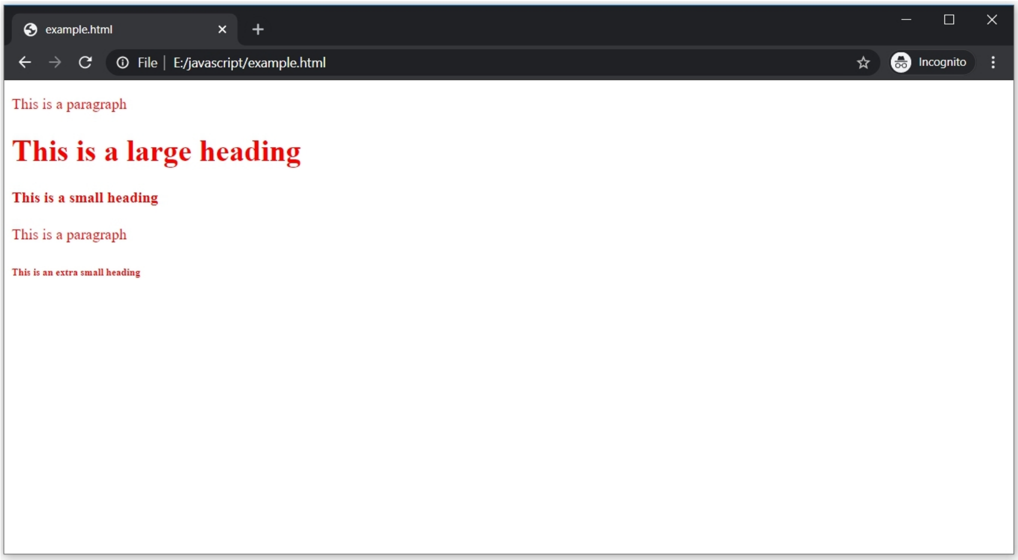
Universal selector

So far, we have discussed how to use the id selector for a particular element, the class selector for multiple elements, and the grouping selector for the grouping different elements with the same definition. But what if we need the same CSS for every element in the HTML file? There is another selector called universal selector for such cases.

Use the asterisk (*) sign to apply CSS to every element.

```html
<!DOCTYPE html>
<html>
    <head>
        <style>
            * {
                color : red;
            }
        </style>


    </head>
    <body>

        <p> This is a paragraph</p>

        <h1> This is a large heading</h1>

        <h4> This is a small heading</h4>

        <p> This is a paragraph</p>

        <h6> This is an extra small heading</h6>


    </body>

</html>
```

The universal selector will apply CSS to every element.



Summary

- CSS selectors are used for finding HTML elements to apply CSS on them
- There are several types of CSS selectors. The most commonly used are: id, class, group, and universal.
- The id selector uses the # (hash) sign with the value of id attribute to select an element. As the id is always unique in an HTML document, this selector is used to apply CSS on a single element.
- The class selector uses the . (dot) with the value of the class attribute to select elements. As multiple elements can have the same class, so this selector is used to apply CSS to multiple elements.
- The grouping selector is used to apply CSS on a group of elements with the same definition.
- The CSS defined with a universal selector is applied to all the elements in an HTML document.

Chapter 14: CSS text and font

Without CSS, the text is boring. CSS has several properties that can be applied to any kind of text, such as paragraphs and headings.

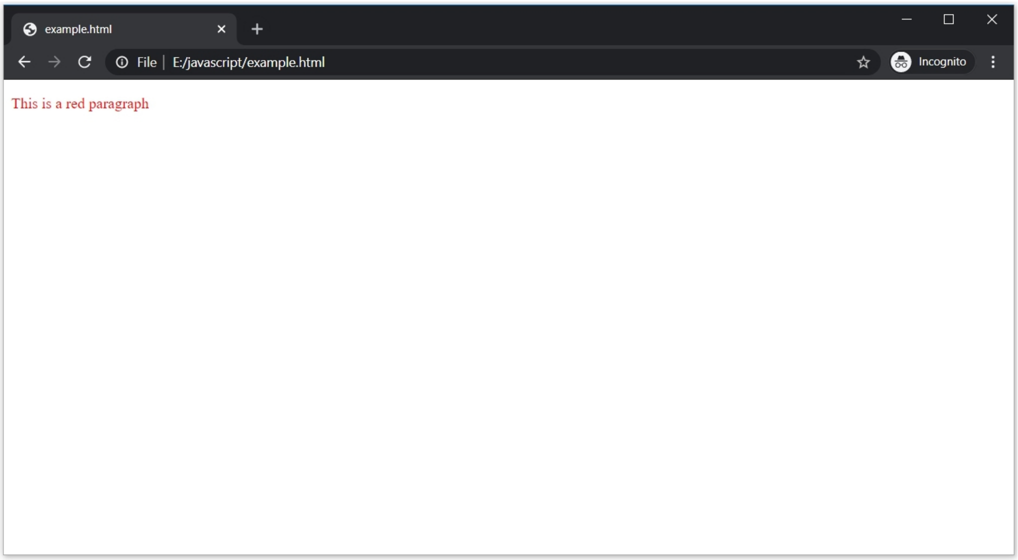
The CSS properties for text are simple, yet very important. We cannot just add boring text on a web page. We need to make enhancements such as **adding color, setting alignment** , etc.

The font is another very useful part. We can add text with different **font styles, sizes, and more using CSS fonts.**

Colors with CSS

In the recent CSS chapters, we saw how text color can be changed using the "color" property. The color property can have **three types of values - color name, HEX value, and RGB value.**

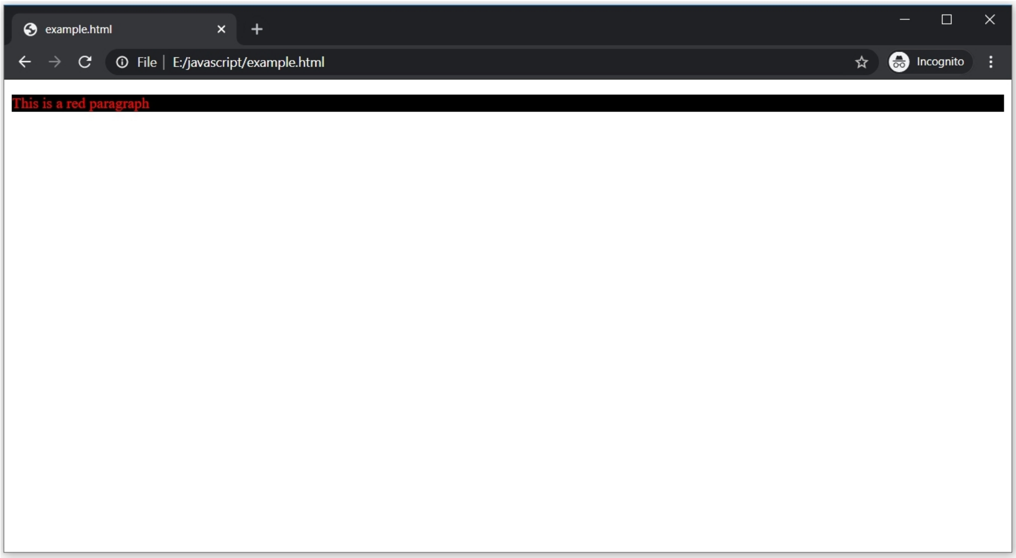
```
<!DOCTYPE html>
<html>
    <head>
        <style>
            p {
                color : red;
            }
        </style>


    </head>
    <body>

        <p> This is a red paragraph</p>

    </body>

</html>
```



We can also change the background color of the text using the **"background-color" property.**

```
<!DOCTYPE html>
<html>
    <head>
        <style>
            p {
                background-color: black;
                color : red;
            }
        </style>


    </head>
    <body>

        <p> This is a red paragraph</p>

    </body>

</html>
```

Similar to the "color" property, the "background-color" can also have the color name, HEX value, or RGB value as its value.



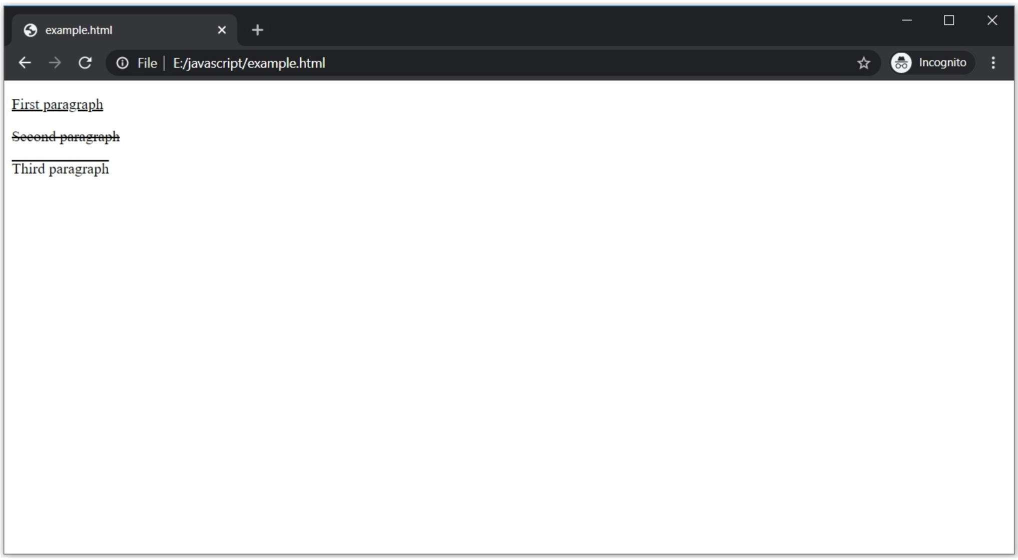
Text decoration

Generally, text in HTML does not have any decorations. To add decorations or to remove the default ones, use the "text-decoration" property.

The "text-decoration" property can have four values: **underline, line-through, overline, and none.** The first three values add decorations while the last one removes the default decoration.

```html
<!DOCTYPE html>
<html>
    <head>
        <style>

            #firstp {
                text-decoration : underline;
            }

            #secondp {
                text-decoration : line-through;
            }

            #thirdp {
                text-decoration : overline;
            }
        </style>


    </head>
    <body>

        <p id="firstp"> First paragraph </p>
        <p id="secondp"> Second paragraph </p>
        <p id="thirdp"> Third paragraph </p>

    </body>

</html>
```

All the three <p> tags have different text decorations - underline for the first, line-through for the second, and overline for the third.

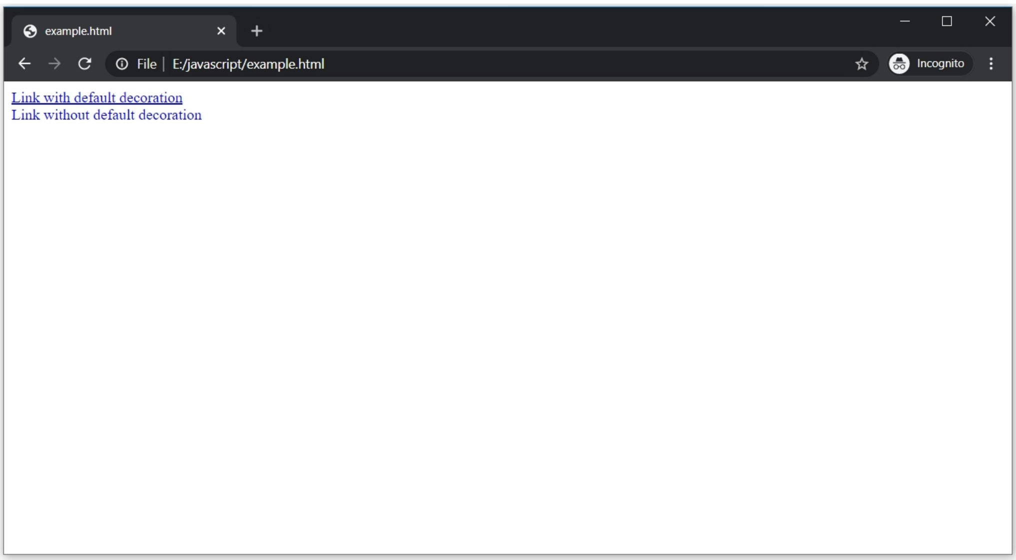


If there is some default decoration, then we can remove it using "none" as the value of "text-decoration".

For example, hyperlinks in HTML are underlined by default. Let's remove the underline using the "text-decoration" property.

```
<!DOCTYPE html>
<html>
    <head>
        <style>

            #seconda {
                text-decoration : none;
            }

        </style>


    </head>
    <body>

        <a href="#" id="firsta"> Link with default decoration </a>
        <br>
        <a href="#" id="seconda"> Link without default decoration </a>
    </body>

</html>
```

Text decoration is given to the second <a> tag while the first one is unchanged.



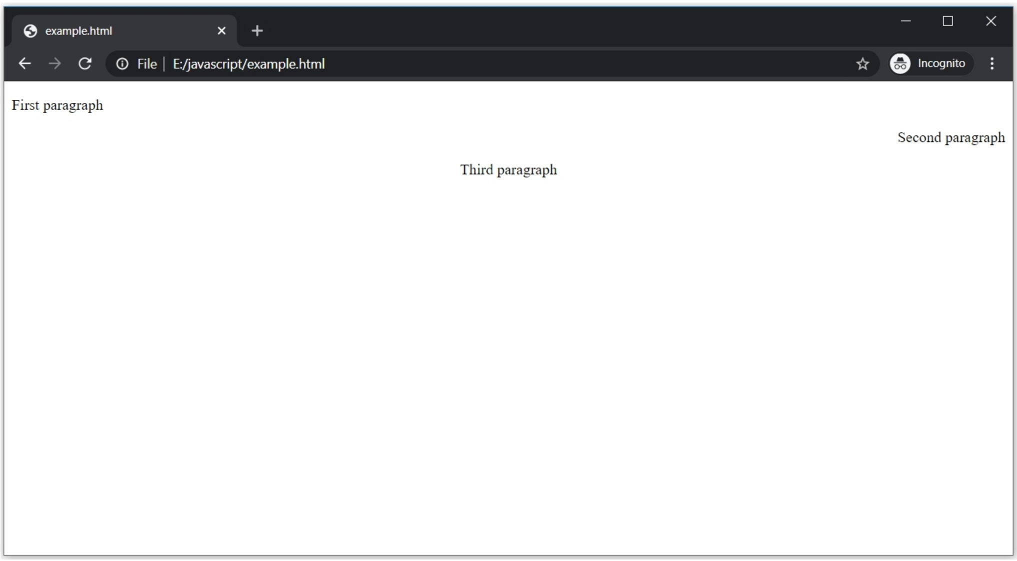
Aligning text

By default, the text is always aligned to the left side of the container. But we can also change the alignment using CSS.

The "text-align" property sets the horizontal alignment. It can have four values: **left, right, and center.**

```
<!DOCTYPE html>
<html>
    <head>
        <style>

            #firstp {
                text-align : left;
            }

            #secondp {
                text-align : right;
            }

            #thirdp {
                text-align : center;
            }
        </style>


    </head>
    <body>

        <p id="firstp"> First paragraph </p>
        <p id="secondp"> Second paragraph </p>
        <p id="thirdp"> Third paragraph </p>

    </body>

</html>
```

The first paragraph is aligned to the left, which is also the default alignment, the second paragraph is aligned to the right, and the third one is aligned to the center.



CSS fonts

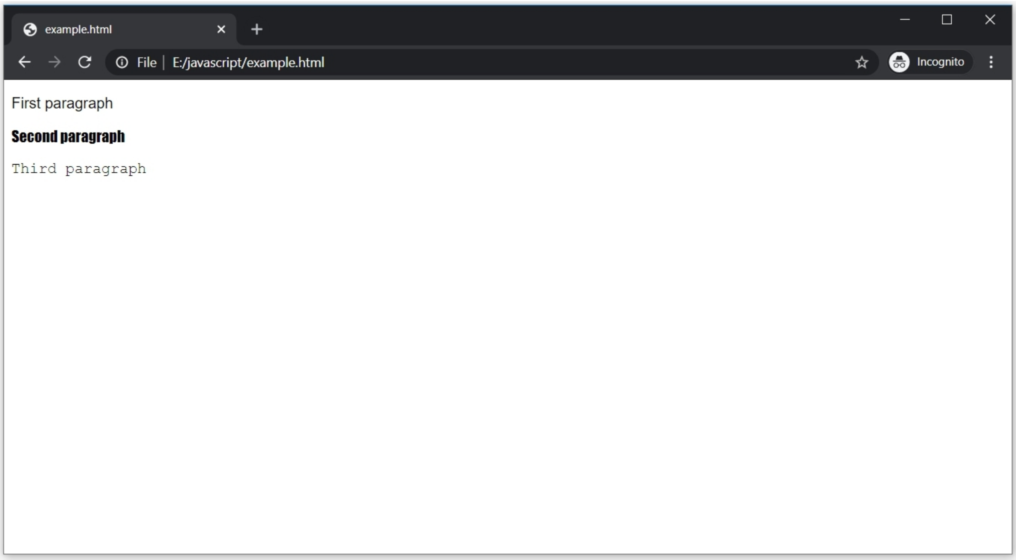
By using CSS font properties, we can change the font family, size, style, and more.

Font family

To change the font family, use the "font-family" property.

```
<!DOCTYPE html>
<html>
    <head>
        <style>

            #firstp {
                font-family: sans-serif;
            }

            #secondp {
                font-family: Impact;
            }

            #thirdp {
                font-family: Courier;
            }
        </style>


    </head>
    <body>

        <p id="firstp"> First paragraph </p>
        <p id="secondp"> Second paragraph </p>
        <p id="thirdp"> Third paragraph </p>

    </body>

</html>
```

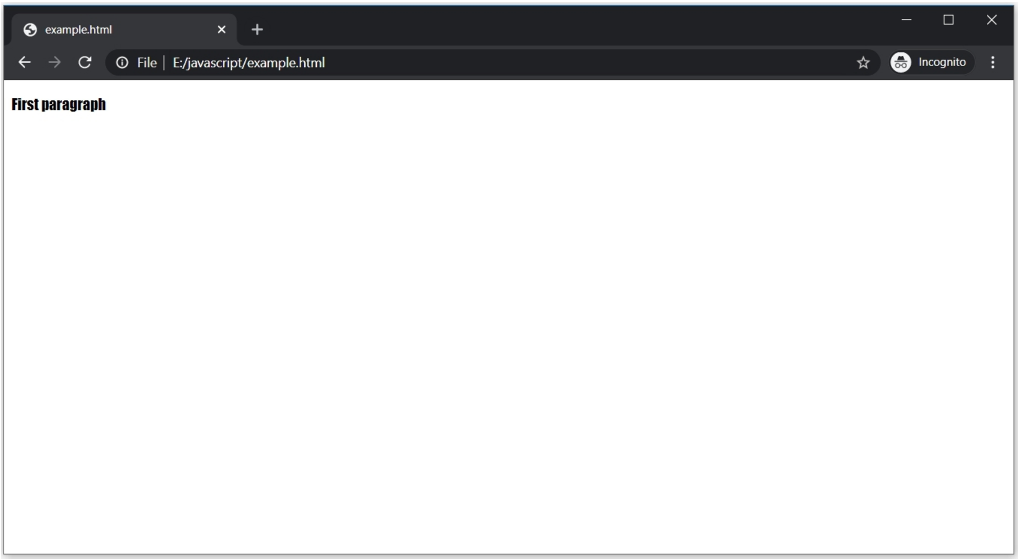
sans-serif, Impact, and Courier are some of the most commonly used fonts in CSS.



Suppose, the font we applied is not supported by the browser, then what? Simply **add some extra fonts separated by commas if the prior ones are not supported by the browser.**

```
<!DOCTYPE html>
<html>
    <head>
        <style>

            #firstp {
                font-family: xxxx, Impact;
            }

        </style>


    </head>
    <body>

        <p id="firstp"> First paragraph </p>

    </body>

</html>
```

xxxx is not a font so the browser will not support it. The next font, Impact, separated by a comma will be assigned as the value of the font-family.

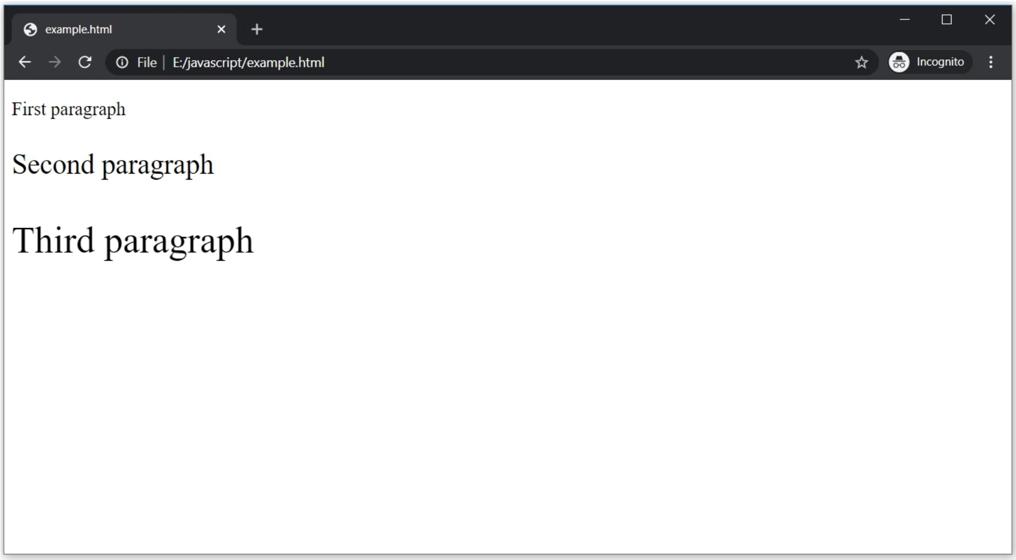


Font size

The default size of a <p> tag is 16px (1em). To change the size, use the "font-size" property. The value can be in pixels, Em, or percentage.

```
<!DOCTYPE html>
<html>
    <head>
        <style>

            #firstp {
                font-size : 20px;
            }

            #secondp {
                font-size : 30px;
            }

            #thirdp {
                font-size : 40px;
            }
        </style>


    </head>
    <body>

        <p id="firstp"> First paragraph </p>
        <p id="secondp"> Second paragraph </p>
        <p id="thirdp"> Third paragraph </p>

    </body>

</html>
```

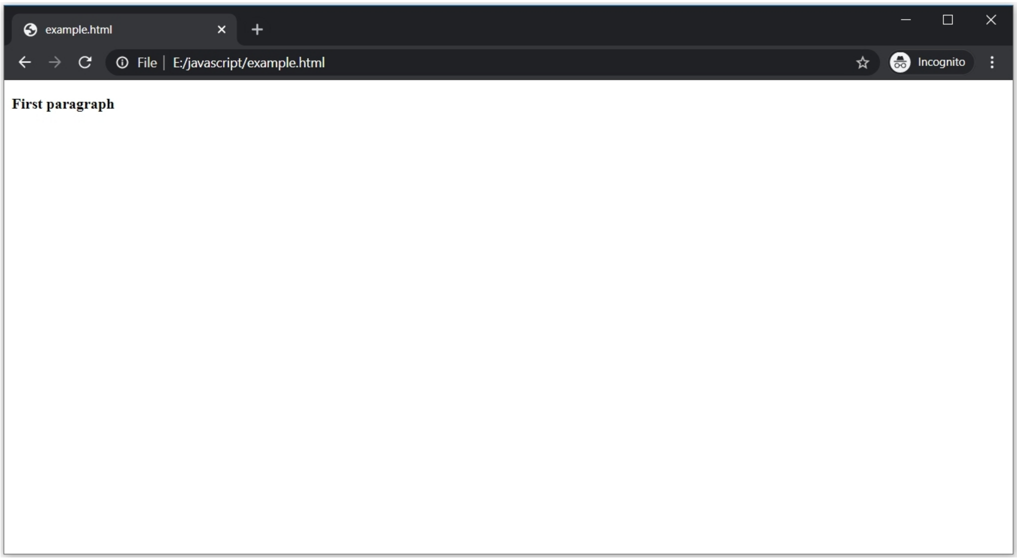
Every <p> tag has a different font size.



Bold font

To make the font bold, use the "font-weight" property.

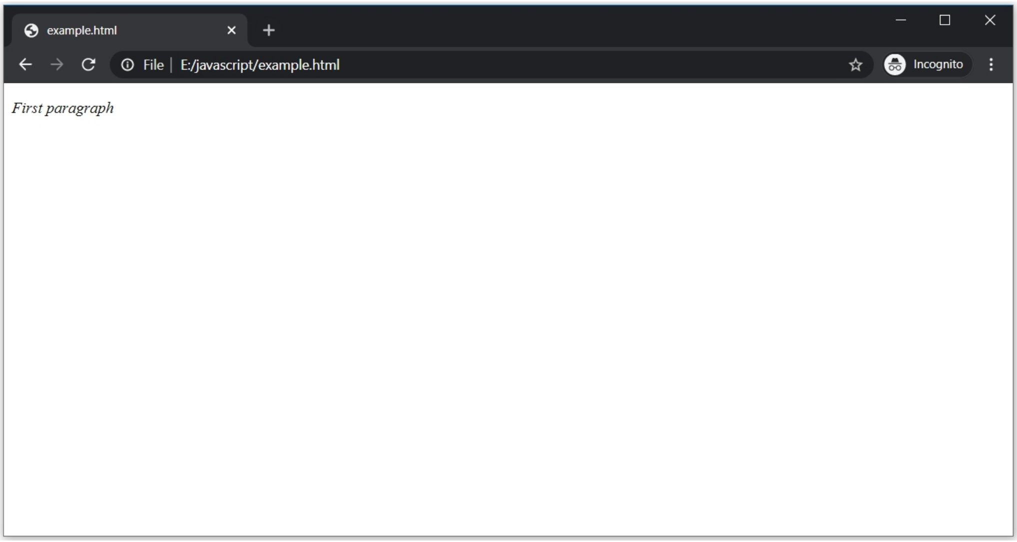
```
<!DOCTYPE html>
<html>
    <head>
        <style>

            p {
                font-weight : bold;
            }

        </style>


    </head>
    <body>

        <p> First paragraph </p>

    </body>

</html>
```



Italic font

To make the font italic, use the "font-style" property.

```
<!DOCTYPE html>
<html>
    <head>
        <style>

            p {
                font-style: italic;
            }

        </style>


    </head>
    <body>

        <p> First paragraph </p>

    </body>

</html>
```



Summary

- The color property is used to change the color of the text.
- To change the background of the text, use the background-color property.
- The text-decoration property is used to add or remove decorations. It can have four values: underline, overline, line-through, and none.
- The text-align property is used to align the text horizontally.

Chapter 15 - CSS borders, margin, and padding

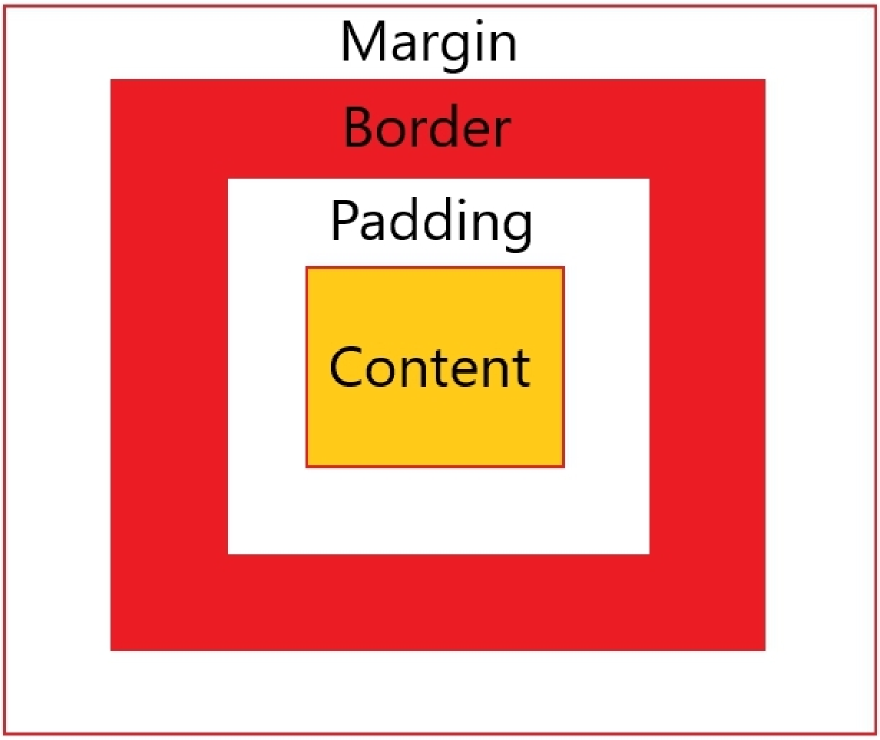
Every HTML element has a border, margin, and padding. These three terms are important because the presentation generally somewhat depends on them.

The border, margin, and padding is used for clarity and better looks. But before moving to them, you need to understand the CSS box model.

CSS box model

The HTML elements are **enclosed inside a box** . This box is knowns as the "CSS Box Model".

This model has four parts: the border, margin, padding, and content. **By default, only content is visible while the border is not and the margin and padding are zero.**



- The content can be text, image, etc.
- The content is surrounded by the border.

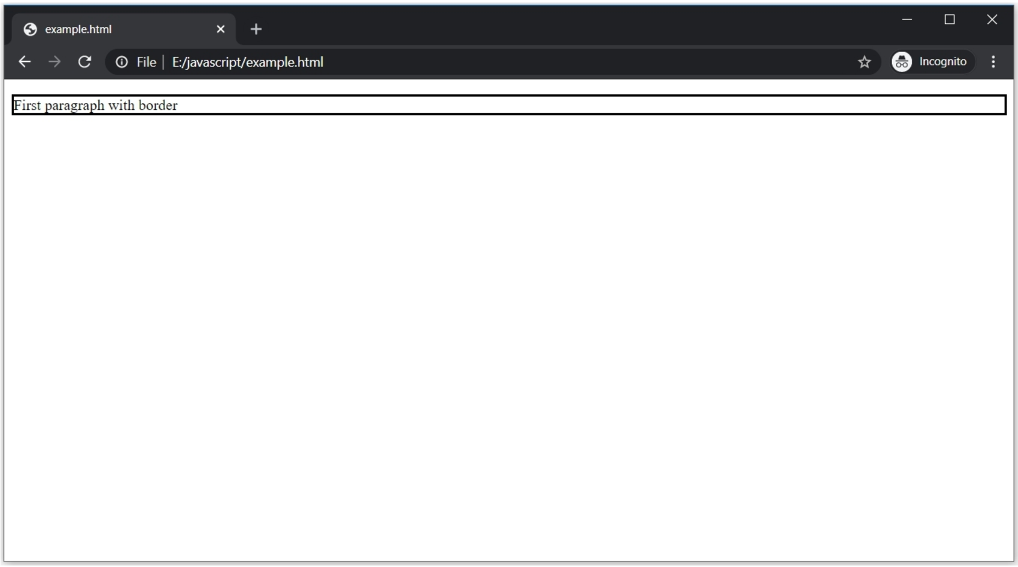
- The area between the content and the border is padding.
- The area outside the border is the margin.
- The border can be visible, depends on the CSS. But margin and padding are invisible.

Border

So as mentioned above, by default, there is no border for HTML elements. **The "border-width" property is used to define the width of the border.** Its value can be in px, cm, pt, em, or the three pre-defined values: thin, thick, or medium. Along with border-width, we also need to **define the style of the border using the "border-style" property.** The "border-style" property can have values such as solid, dotted, and dashed.

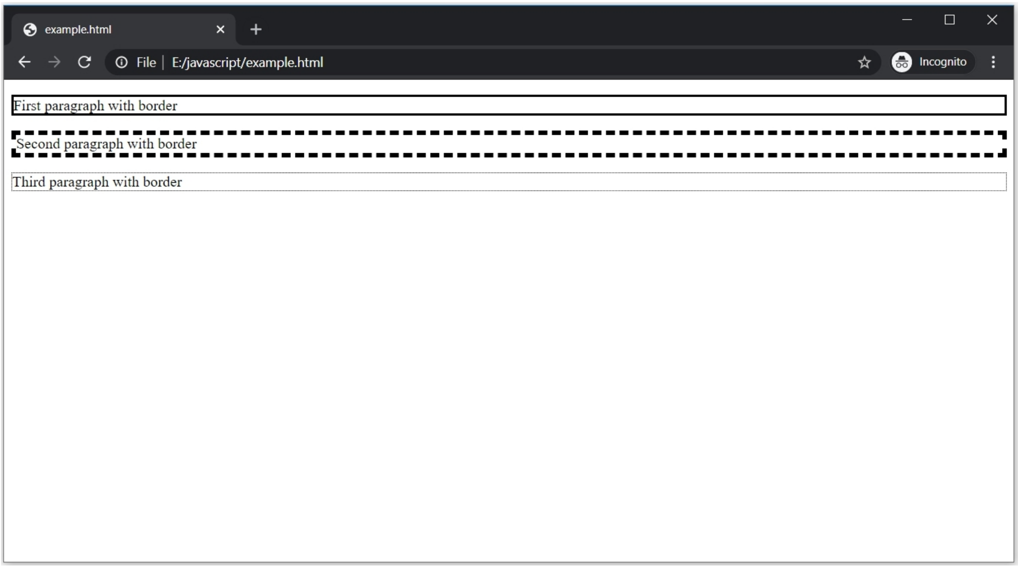
```
<!DOCTYPE html>
<html>
    <head>
        <style>

          #firstp {
              border-width: 2px;
              border-style: solid;
          }

        </style>
    </head>
    <body>

        <p id="firstp"> First paragraph with border </p>

    </body>

</html>
```

The <p> tag has "border-width" of 2px and solid "border-style".



Let's try some other variants.

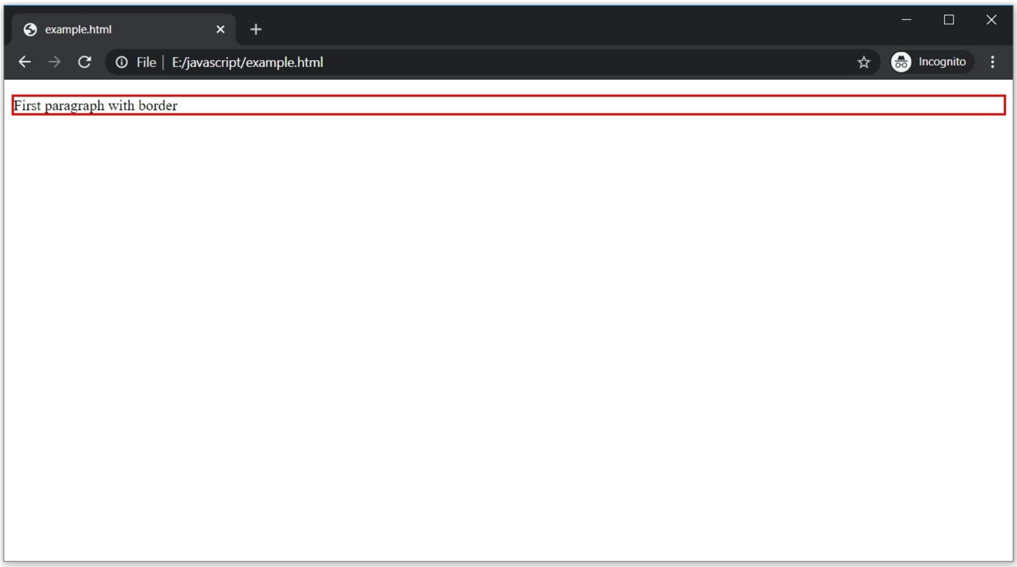
```
<!DOCTYPE html>
<html>
    <head>
        <style>

            #firstp {
                border-width: 2px;
                border-style: solid;
            }

            #secondp {
                border-width: 5px;
                border-style: dashed;
            }

            #thirdp {
                border-width: thin;
                border-style: dotted;
            }

        </style>
    </head>
    <body>

        <p id="firstp"> First paragraph with border </p>

        <p id="secondp"> Second paragraph with border </p>

        <p id="thirdp"> Third paragraph with border </p>

    </body>

</html>
```



By default, the color of the border is black. But it can be **changed using the "border-color" property.**

```
<!DOCTYPE html>
<html>
    <head>
        <style>

            #firstp {
                border-width: 2px;
                border-style: solid;
                border-color: red;
            }

        </style>
    </head>
    <body>

        <p id="firstp"> First paragraph with border </p>

    </body>

</html>
```

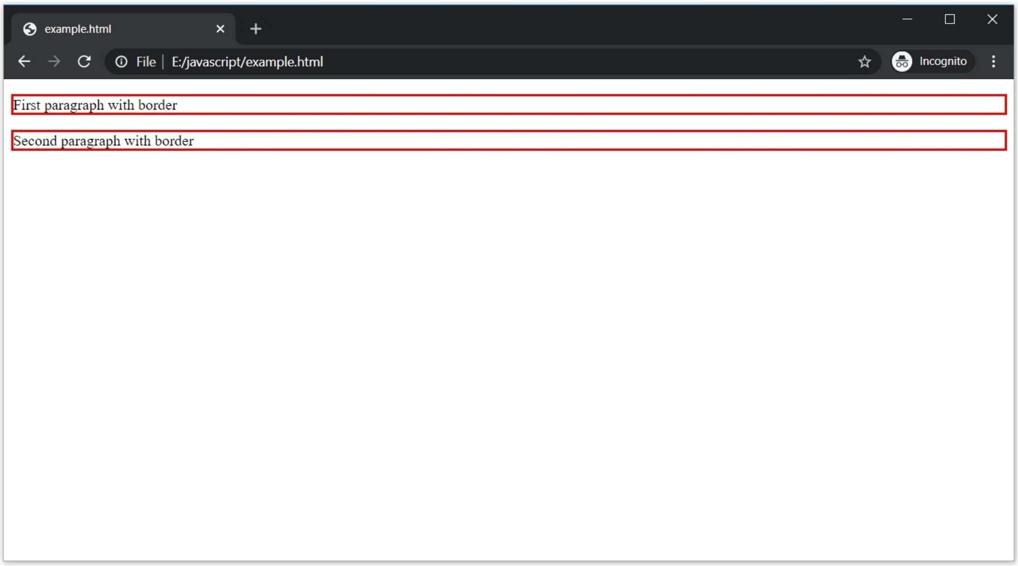
The value of the "border-color" property can be a color name, HEX value, RGB value, HSL value, or transparent.



These three properties deal with borders, but another property is the shorthand property for borders. This property is simply called "border" property and consists of the width, style, and color.

```
<!DOCTYPE html>
<html>
    <head>
        <style>

            #firstp {
                border-width: 2px;
                border-style: solid;
                border-color: red;
            }

            #secondp {
                border: 2px solid red;
            }

        </style>
    </head>
    <body>

        <p id="firstp"> First paragraph with border </p>

        <p id="secondp"> Second paragraph with border </p>

    </body>

</html>
```

The first <p> tag has all the three border properties while the second one has **the shorthand property** with similar CSS.

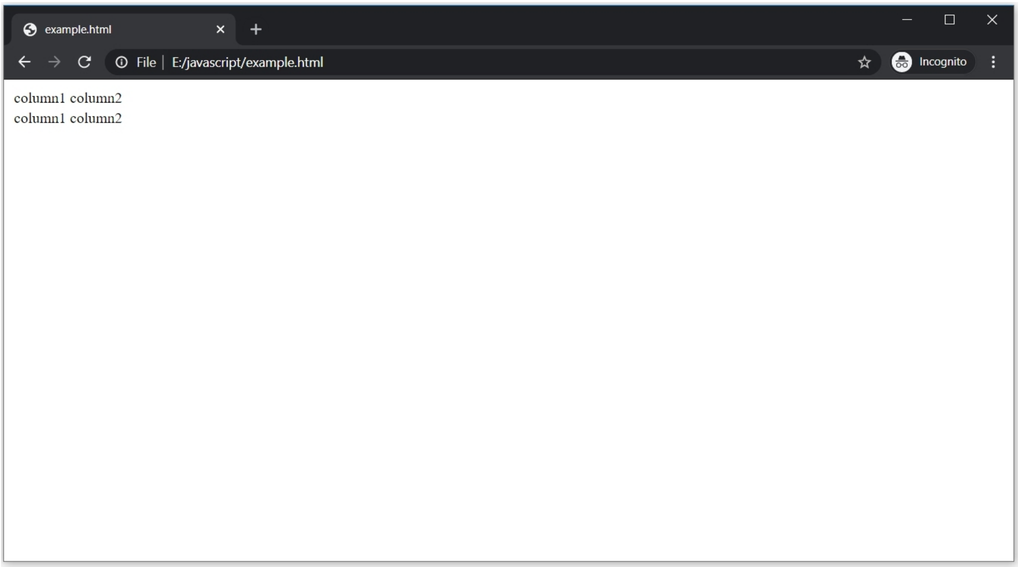


Tables with border

In the HTML section, we discussed how attractive tables can be created using CSS.

```
<!DOCTYPE html>
<html>
    <body>

        <table>
            <tr>
                <td> column1 </td>
                <td> column2 </td>
            </tr>
            <tr>
                <td> column1 </td>
                <td> column2 </td>
            </tr>
        </table>

    </body>
</html>
```

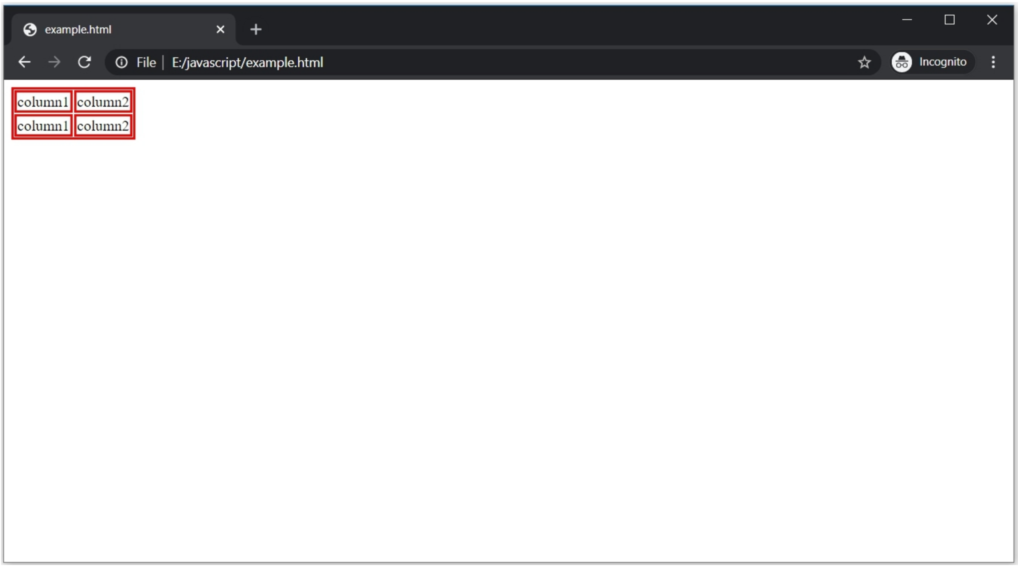
This table does not have any CSS.



To add the border, we can use any property we discussed. Let's go with the "border" property.

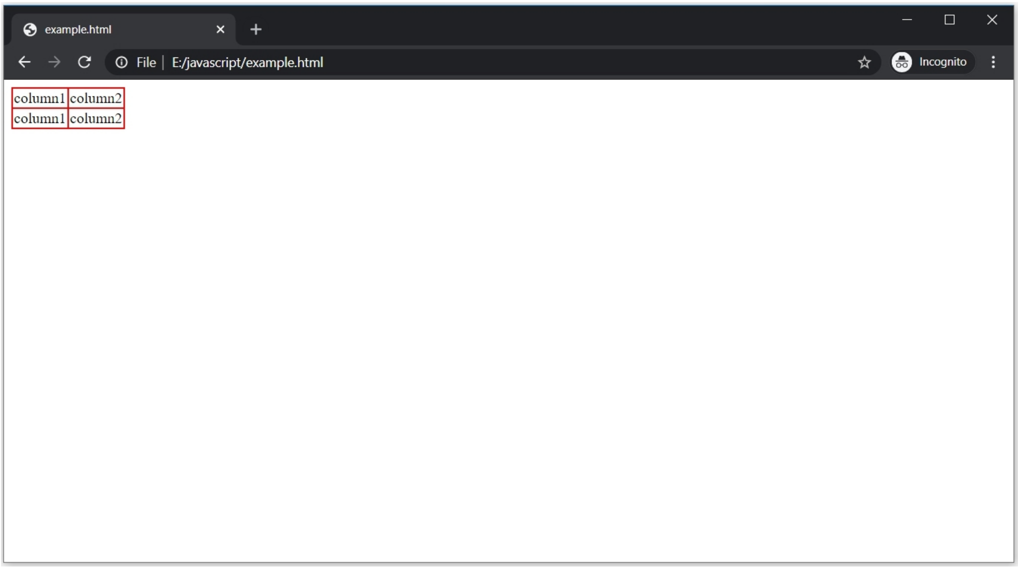
```
<!DOCTYPE html>
<html>
    <head>
        <style>
            table, td{
                border : 2px solid red;
            }
        </style>
    </head>
    <body>

        <table>
            <tr>
                <td> column1 </td>
                <td> column2 </td>
            </tr>
            <tr>
                <td> column1 </td>
                <td> column2 </td>
            </tr>
        </table>

    </body>
</html>
```

To create a proper border on a table, we need to apply CSS to the <table> tag as well as the <td> tag.



As of now, the table has a double border. To fix this, **use the "border-collapse" property and set its value to "collapse".**

```
<!DOCTYPE html>
<html>
    <head>
        <style>
            table, td{
                border : 2px solid red;
                border-collapse : collapse;
            }
        </style>
    </head>
    <body>

        <table>
            <tr>
                <td> column1 </td>
                <td> column2 </td>
            </tr>
            <tr>
                <td> column1 </td>
                <td> column2 </td>
            </tr>
        </table>

    </body>
</html>
```

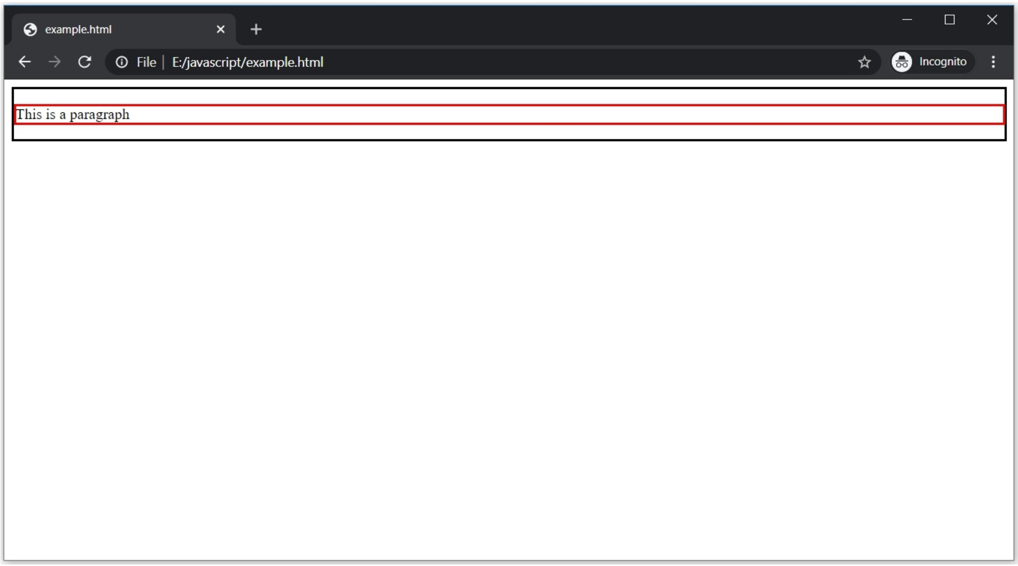


Margin

To understand, we need to create the content inside a <div> tag. This tag is used to define a section or division in an HTML document.

```
<!DOCTYPE html>
<html>
    <head>
        <style>

            div {
                border : 2px solid black;
            }

            p {
                border : 2px solid red;
            }

        </style>
    </head>
    <body>

        <div>
            <p>This is a paragraph</p>
        </div>

    </body>
</html>
```

Both <p> and <div> tags have border of 2px.

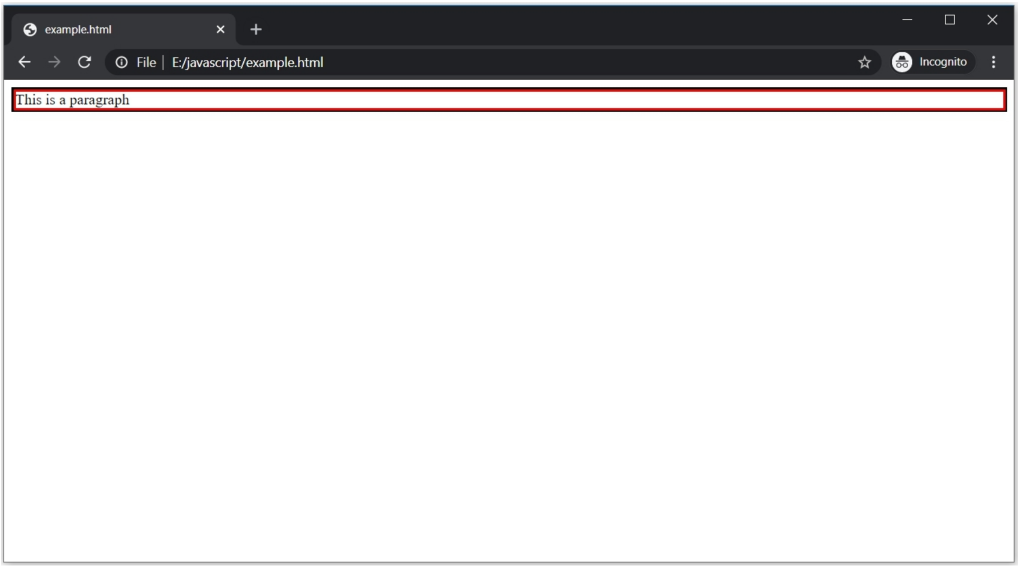


The area between the red line and the black line is the margin for the <p> tag. To set the margin in all the directions, we can use the "margin" property.

Let's remove the default margin by setting its value to 0px.

```
<!DOCTYPE html>
<html>
    <head>
        <style>

            div {
                border : 2px solid black;
            }

            p {
                border : 2px solid red;
                margin : 0px;
            }

        </style>
    </head>
    <body>

        <div>
            <p>This is a paragraph</p>
        </div>

    </body>
</html>
```

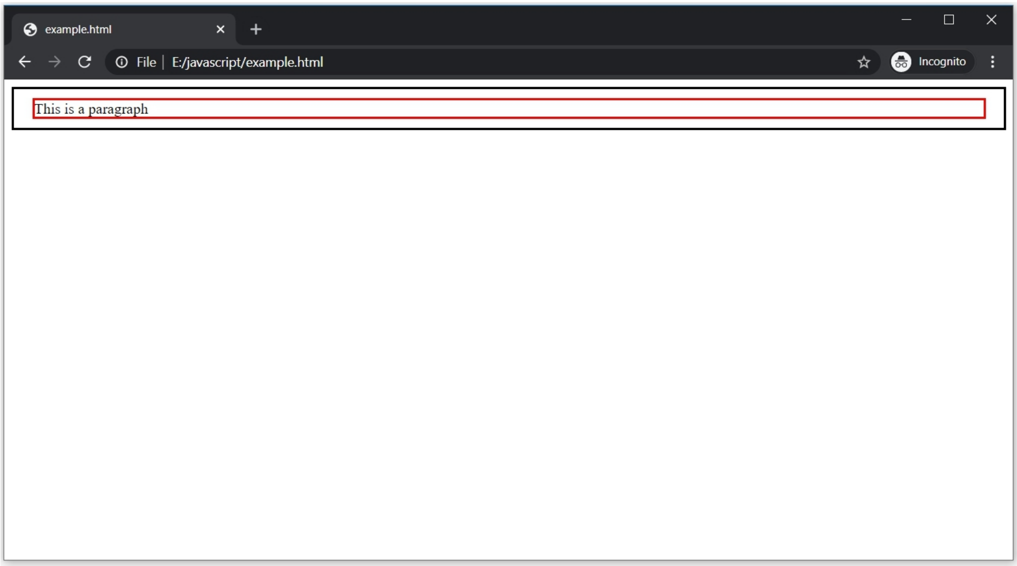
There is no margin now.



To set margin in a particular direction, we have four different properties: **"margin-top", "margin-bottom", "margin-right", and "margin-left".**

```
<!DOCTYPE html>
<html>
    <head>
        <style>

            div {
                border : 2px solid black;
            }

            p {
                border : 2px solid red;
                margin-top : 10px;
                margin-bottom : 10px;
                margin-left : 20px;
                margin-right : 20px;
            }

        </style>
    </head>
    <body>

        <div>
            <p>This is a paragraph</p>
        </div>

    </body>
</html>
```

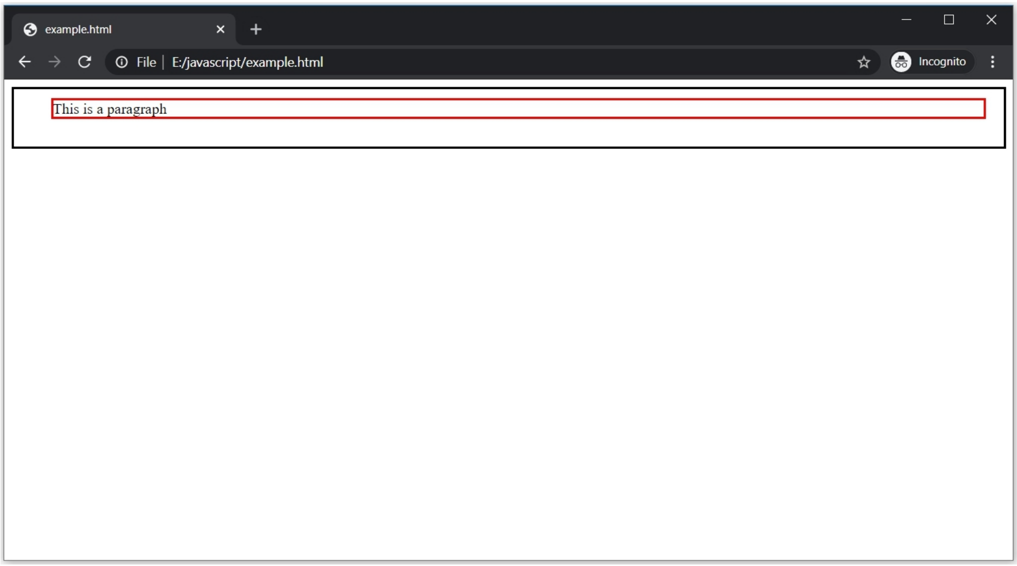
Margin for top and bottom is 10px while it is 20px for left and right.



The "margin" property is actually a shorthand property. We can also use the "margin" property to set different margins for each direction.

```
<!DOCTYPE html>
<html>
    <head>
        <style>

            div {
                border : 2px solid black;
            }

            p {
                border : 2px solid red;
                margin : 10px 20px 30px 40px;
            }

        </style>
    </head>
    <body>

        <div>
            <p>This is a paragraph</p>
        </div>

    </body>
</html>
```

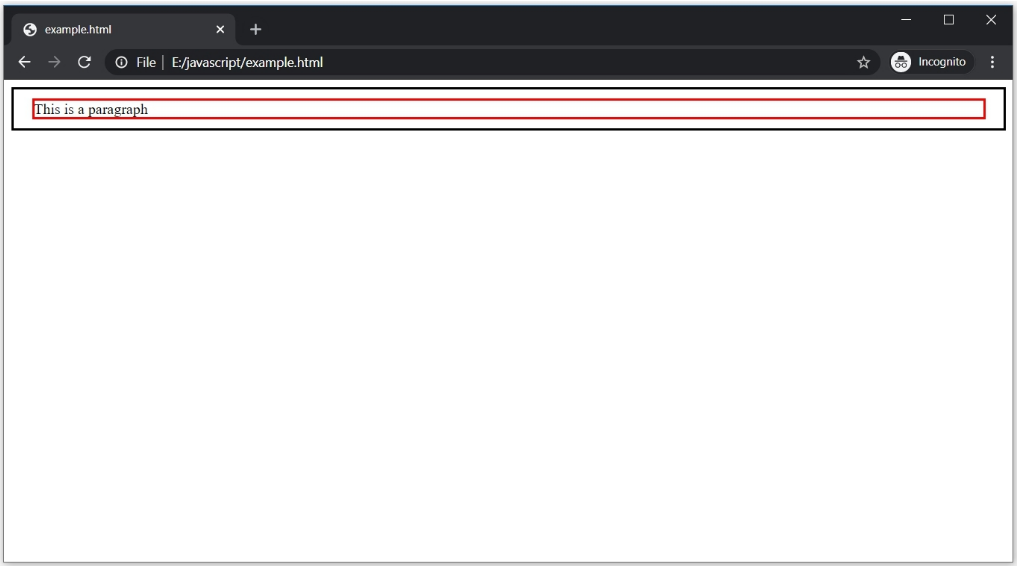
Here, margin top is 10px, right is 20px, bottom is 30px, and left is 40px.



Another way to use the margin property is by giving only two values.

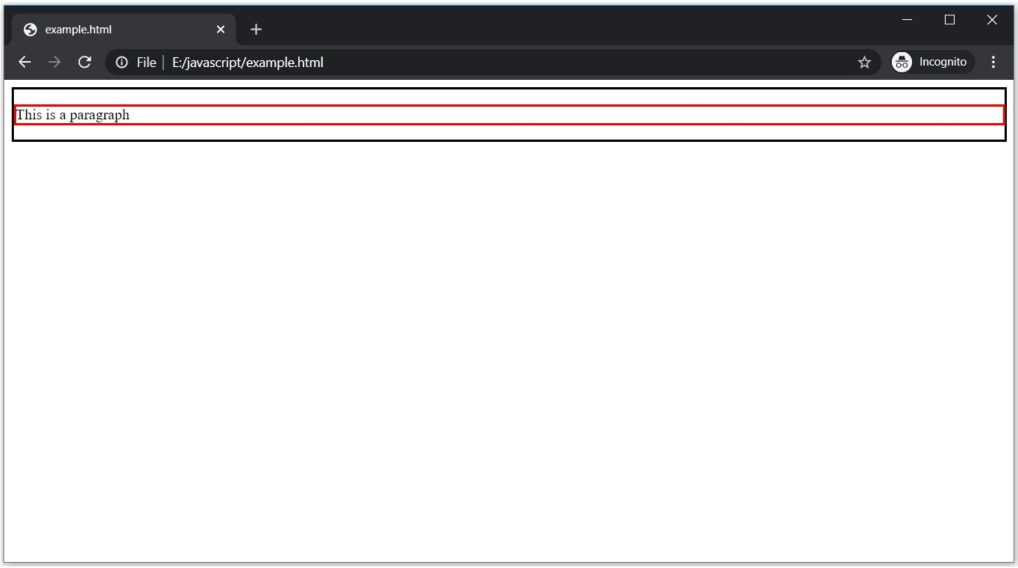
```
<!DOCTYPE html>
<html>
    <head>
        <style>

            div {
                border : 2px solid black;
            }

            p {
                border : 2px solid red;
                margin : 10px 20px;
            }

        </style>
    </head>
    <body>

        <div>
            <p>This is a paragraph</p>
        </div>

    </body>
</html>
```

Here, the value for the top and bottom margin is 10px while the value for right and left is 20px.



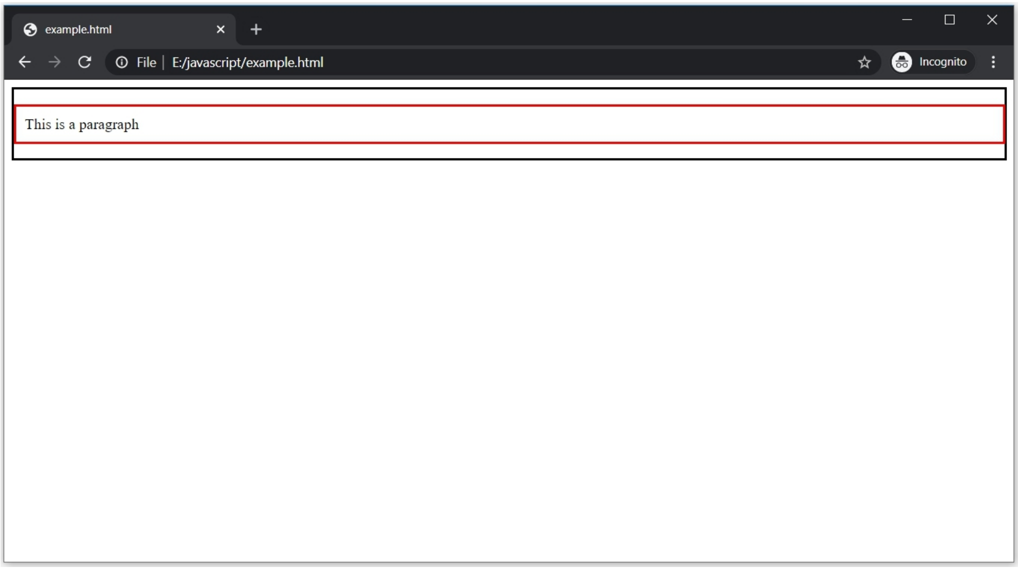
Padding

The area between the content and the red line is the padding for the <p> tag.



The "padding" property is used to change the padding.

```html
<!DOCTYPE html>
<html>
    <head>
        <style>

            div {
                border : 2px solid black;
            }

            p {
                border : 2px solid red;
                padding: 10px;
            }

        </style>
    </head>
    <body>

        <div>
            <p>This is a paragraph</p>
        </div>

    </body>
</html>
```



For padding, we also have properties like "padding-top", "padding-bottom", "padding-right", and "padding-left". Moreover, the "padding" property is also a shorthand property like the "margin" property.

Summary

- The CSS box model consists of the border, margin, and padding.
- The border can be set using the "border" shorthand property or properties like "border-width", "border-style", "border-color".

- The "margin" property is a shorthand property used to set margin in various ways. To set margin for particular directions, use "margin-top", "margin-bottom", "margin-left", or "margin-right".
- The "padding" property is a shorthand property used to set padding in various ways. To set padding for particular directions, use "padding-top", "padding-bottom", "padding-left", or "padding-right".

Chapter 16 - CSS backgrounds

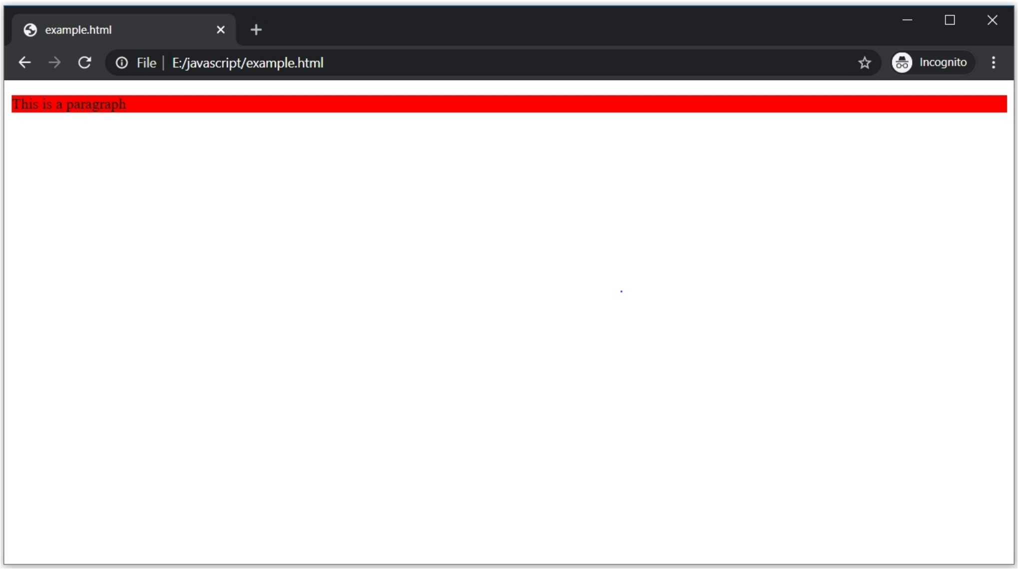
By default, every HTML page has a white background. It is boring. There are a few CSS properties that can be used to manipulate backgrounds. We can **change the background color, or add an image to the background.**

Background color

We can change the background color of any HTML element **using the "background-color" property.**

```
<!DOCTYPE html>
<html>
    <head>
        <style>

            p {
                background-color : red;
            }

        </style>
    </head>
    <body>

        <p>This is a paragraph</p>

    </body>
</html>
```

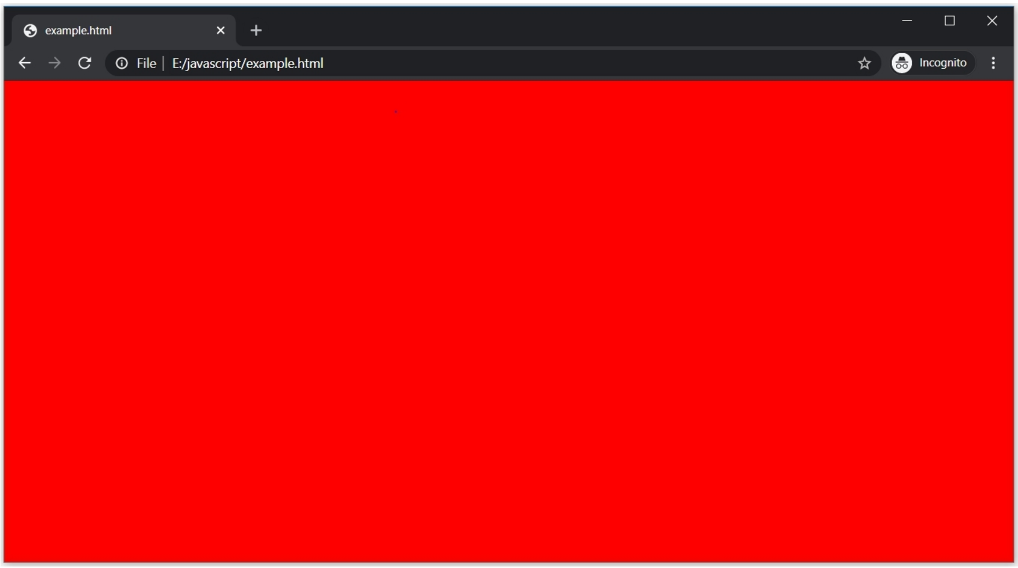
The <p> tag has a red background color.



But generally, "background-color" property is used to set the background of the entire page. To do this, we have to add the "background-color" property to the body of the page.

```
<!DOCTYPE html>
<html>
    <head>
        <style>

            body {
                background-color : red;
            }

        </style>
    </head>
    <body>


    </body>
</html>
```

Remember, any CSS added to the body will affect the entire HTML page.



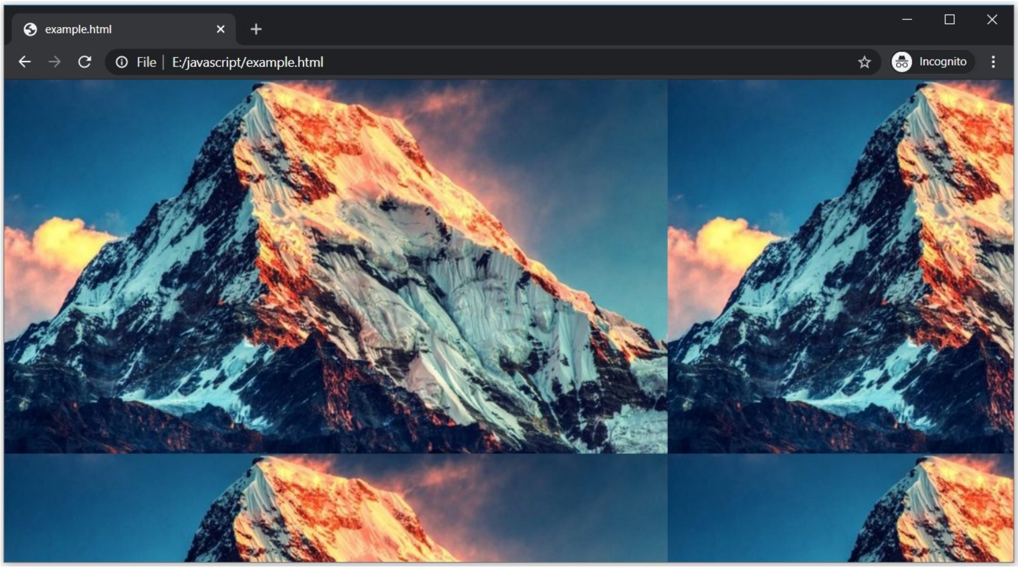
Other than the color name, we can also assign HEX value or RGB value as the value of "background-color" property.

Background image

Apart from color, we can also set an image as the background using the "background-color" property. **Use the url() method as its value and pass the link of the image as an argument.**

```
<!DOCTYPE html>
<html>
    <head>
        <style>

            body {
                background-image :url("./images/mountain.jpg");
            }

        </style>
    </head>
    <body>


    </body>
</html>
```

If you are uploading from your machine, then you have to give the exact path or if it an image from the internet, give its proper url.



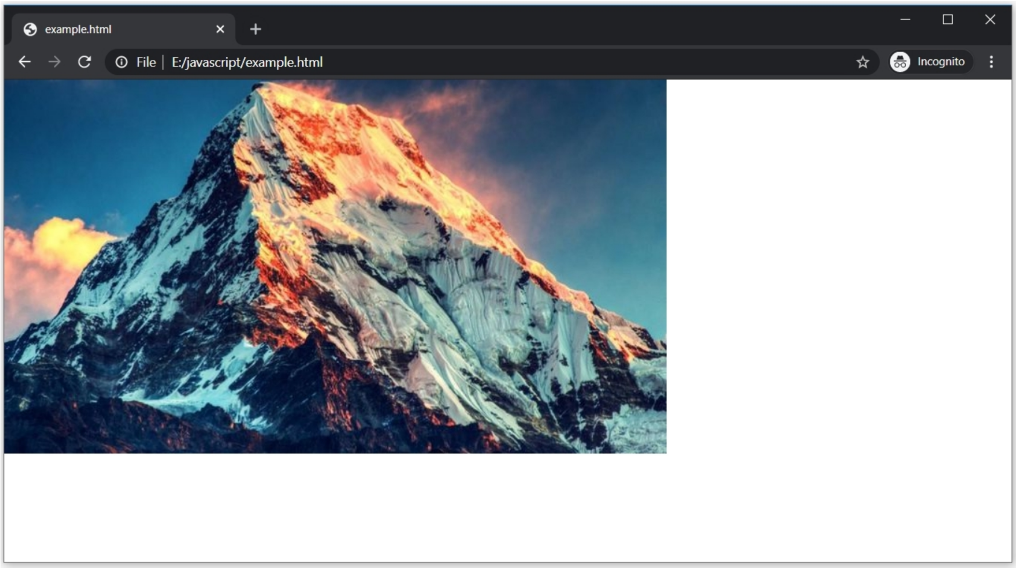
The image appears in the background but it is not in the proper way. To adjust the image properly, we have additional CSS properties.

Background repeat

In the above example, the background image is repeating. This happens because the image is set according to its actual size.

To prevent the image from repeating, **use the "background-repeat" property and set "no-repeat" as its value.**

```
<!DOCTYPE html>
<html>
    <head>
        <style>

            body {
                background-image : url("./images/mountain.jpg");
                background-repeat : no-repeat;
            }

        </style>
    </head>
    <body>


    </body>
</html>
```



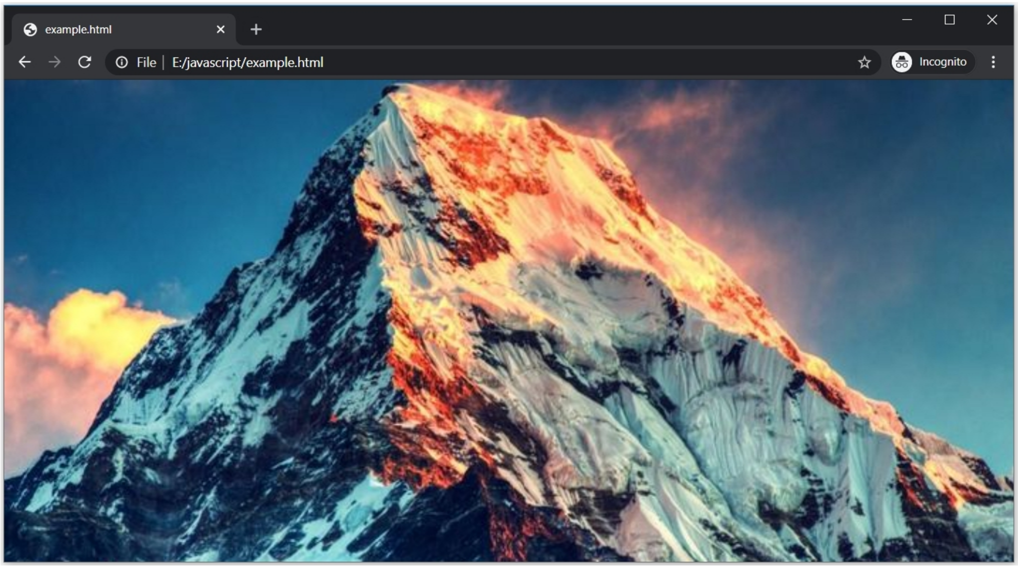
The background image is not repeating now.

Apart from "no-repeat", we can use "repeat-x" and "repeat-y" to repeat the background horizontally and vertically, respectively.

Background cover

Still, the background image does not appear property. It should cover the whole window and to do this, use the "background-size" property and set "cover" as its value.

```html
<!DOCTYPE html>
<html>
    <head>
        <style>

            body {
                background-image : url("./images/mountain.jpg");
                background-repeat : no-repeat;
                background-size : cover;
            }

        </style>
    </head>
    <body>


    </body>
</html>
```



This looks perfect now.

There are other properties such as "background-attachment" that is used to specify if the image is fixed or scrollable and "background-position" to set the position of the image.

Summary

- The "background-color" property is used to set the color of the background.
- The "background-image" property is used to set an image as the background.
- To adjust the repetition of the background image, use the "background-repeat" property. Set its value as "no-repeat" if no repetition is required.
- To stretch the image to the whole window, use the "background-size" property and set its value as "cover".